SG No. 191 D_{6h}^{1} $P6/mmm$ [hexagonal]

* plus set: $+[0, 0, 0]$

* Wyckoff site: `1a`, site symmetry: `6/mmm`

Table 1: Wyckoff bond: `1a@1a`, bond symmetry: `6/mmm`

No.	vector	center	mapping
1	[0, 0, Z]	[0, 0, 0]	`[1,2,3,4,5,6,-7,-8,-9,-10,-11,-12,-13,-14,-15,-16,-17,-18,19,20,21,22,23,24]`

Table 2: Wyckoff bond: `3b@1a`, bond symmetry: `mmm`

No.	vector	center	mapping
1	[X, $2X$, 0]	[0, 0, 0]	`[1,-4,-8,11,-13,16,20,-23]`
2	[$-2X$, $-X$, 0]	[0, 0, 0]	`[2,-5,-7,10,-14,17,19,-22]`
3	[X, $-X$, 0]	[0, 0, 0]	`[3,-6,-9,12,-15,18,21,-24]`

Table 3: Wyckoff bond: `3c@1a`, bond symmetry: `mmm`

No.	vector	center	mapping
1	[X, 0, 0]	[0, 0, 0]	`[1,-4,8,-11,-13,16,-20,23]`
2	[0, X, 0]	[0, 0, 0]	`[2,-5,7,-10,-14,17,-19,22]`
3	[$-X$, $-X$, 0]	[0, 0, 0]	`[3,-6,9,-12,-15,18,-21,24]`

Table 4: Wyckoff bond: `6d@1a`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	[X, Y, 0]	[0, 0, 0]	`[1,-4,-13,16]`
2	[$-Y$, $X-Y$, 0]	[0, 0, 0]	`[2,-5,-14,17]`
3	[$-X+Y$, $-X$, 0]	[0, 0, 0]	`[3,-6,-15,18]`
4	[Y, X, 0]	[0, 0, 0]	`[7,-10,-19,22]`
5	[$X-Y$, $-Y$, 0]	[0, 0, 0]	`[8,-11,-20,23]`
6	[$-X$, $-X+Y$, 0]	[0, 0, 0]	`[9,-12,-21,24]`

Table 5: Wyckoff bond: `6e@1a`, bond symmetry: `.2/m.`

No.	vector	center	mapping
1	[X, $2X$, Z]	[0, 0, 0]	`[1,-8,-13,20]`
2	[$-2X$, $-X$, Z]	[0, 0, 0]	`[2,-7,-14,19]`
3	[X, $-X$, Z]	[0, 0, 0]	`[3,-9,-15,21]`
4	[$-X$, $-2X$, Z]	[0, 0, 0]	`[4,-11,-16,23]`
5	[$2X$, X, Z]	[0, 0, 0]	`[5,-10,-17,22]`
6	[$-X$, X, Z]	[0, 0, 0]	`[6,-12,-18,24]`

Table 6: Wyckoff bond: `6f@1a`, bond symmetry: `..2/m`

No.	vector	center	mapping
1	$[X, 0, Z]$	$[0, 0, 0]$	`[1,-11,-13,23]`
2	$[0, X, Z]$	$[0, 0, 0]$	`[2,-10,-14,22]`
3	$[-X, -X, Z]$	$[0, 0, 0]$	`[3,-12,-15,24]`
4	$[-X, 0, Z]$	$[0, 0, 0]$	`[4,-8,-16,20]`
5	$[0, -X, Z]$	$[0, 0, 0]$	`[5,-7,-17,19]`
6	$[X, X, Z]$	$[0, 0, 0]$	`[6,-9,-18,21]`

Table 7: Wyckoff bond: `12g@1a`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, 0]$	`[1,-13]`
2	$[-Y, X-Y, Z]$	$[0, 0, 0]$	`[2,-14]`
3	$[-X+Y, -X, Z]$	$[0, 0, 0]$	`[3,-15]`
4	$[-X, -Y, Z]$	$[0, 0, 0]$	`[4,-16]`
5	$[Y, -X+Y, Z]$	$[0, 0, 0]$	`[5,-17]`
6	$[X-Y, X, Z]$	$[0, 0, 0]$	`[6,-18]`
7	$[Y, X, -Z]$	$[0, 0, 0]$	`[7,-19]`
8	$[X-Y, -Y, -Z]$	$[0, 0, 0]$	`[8,-20]`
9	$[-X, -X+Y, -Z]$	$[0, 0, 0]$	`[9,-21]`
10	$[-Y, -X, -Z]$	$[0, 0, 0]$	`[10,-22]`
11	$[-X+Y, Y, -Z]$	$[0, 0, 0]$	`[11,-23]`
12	$[X, X-Y, -Z]$	$[0, 0, 0]$	`[12,-24]`

* Wyckoff site: `1b`, site symmetry: `6/mmm`

Table 8: Wyckoff bond: `1a@1b`, bond symmetry: `6/mmm`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,2,3,4,5,6,-7,-8,-9,-10,-11,-12,-13,-14,-15,-16,-17,-18,19,20,21,22,23,24]`

Table 9: Wyckoff bond: `3b@1b`, bond symmetry: `mmm`

No.	vector	center	mapping
1	$[X, 2X, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,-4,-8,11,-13,16,20,-23]`
2	$[-2X, -X, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[2,-5,-7,10,-14,17,19,-22]`
3	$[X, -X, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[3,-6,-9,12,-15,18,21,-24]`

Table 10: Wyckoff bond: `3c@1b`, bond symmetry: `mmm`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,-4,8,-11,-13,16,-20,23]`
2	$[0, X, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[2,-5,7,-10,-14,17,-19,22]`
3	$[-X, -X, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[3,-6,9,-12,-15,18,-21,24]`

Table 11: Wyckoff bond: `6d@1b`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[0, 0, \frac{1}{2}]$	`[1,-4,-13,16]`
2	$[-Y, X-Y, 0]$	$[0, 0, \frac{1}{2}]$	`[2,-5,-14,17]`
3	$[-X+Y, -X, 0]$	$[0, 0, \frac{1}{2}]$	`[3,-6,-15,18]`
4	$[Y, X, 0]$	$[0, 0, \frac{1}{2}]$	`[7,-10,-19,22]`
5	$[X-Y, -Y, 0]$	$[0, 0, \frac{1}{2}]$	`[8,-11,-20,23]`
6	$[-X, -X+Y, 0]$	$[0, 0, \frac{1}{2}]$	`[9,-12,-21,24]`

Table 12: Wyckoff bond: `6e@1b`, bond symmetry: `.2/m.`

No.	vector	center	mapping
1	$[X, 2X, Z]$	$[0, 0, \frac{1}{2}]$	`[1,-8,-13,20]`
2	$[-2X, -X, Z]$	$[0, 0, \frac{1}{2}]$	`[2,-7,-14,19]`
3	$[X, -X, Z]$	$[0, 0, \frac{1}{2}]$	`[3,-9,-15,21]`
4	$[-X, -2X, Z]$	$[0, 0, \frac{1}{2}]$	`[4,-11,-16,23]`
5	$[2X, X, Z]$	$[0, 0, \frac{1}{2}]$	`[5,-10,-17,22]`
6	$[-X, X, Z]$	$[0, 0, \frac{1}{2}]$	`[6,-12,-18,24]`

Table 13: Wyckoff bond: `6f@1b`, bond symmetry: `..2/m`

No.	vector	center	mapping
1	$[X, 0, Z]$	$[0, 0, \frac{1}{2}]$	`[1,-11,-13,23]`
2	$[0, X, Z]$	$[0, 0, \frac{1}{2}]$	`[2,-10,-14,22]`
3	$[-X, -X, Z]$	$[0, 0, \frac{1}{2}]$	`[3,-12,-15,24]`
4	$[-X, 0, Z]$	$[0, 0, \frac{1}{2}]$	`[4,-8,-16,20]`
5	$[0, -X, Z]$	$[0, 0, \frac{1}{2}]$	`[5,-7,-17,19]`
6	$[X, X, Z]$	$[0, 0, \frac{1}{2}]$	`[6,-9,-18,21]`

Table 14: Wyckoff bond: `12g@1b`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, \frac{1}{2}]$	`[1,-13]`
2	$[-Y, X-Y, Z]$	$[0, 0, \frac{1}{2}]$	`[2,-14]`
3	$[-X+Y, -X, Z]$	$[0, 0, \frac{1}{2}]$	`[3,-15]`
4	$[-X, -Y, Z]$	$[0, 0, \frac{1}{2}]$	`[4,-16]`
5	$[Y, -X+Y, Z]$	$[0, 0, \frac{1}{2}]$	`[5,-17]`
6	$[X-Y, X, Z]$	$[0, 0, \frac{1}{2}]$	`[6,-18]`
7	$[Y, X, -Z]$	$[0, 0, \frac{1}{2}]$	`[7,-19]`
8	$[X-Y, -Y, -Z]$	$[0, 0, \frac{1}{2}]$	`[8,-20]`
9	$[-X, -X+Y, -Z]$	$[0, 0, \frac{1}{2}]$	`[9,-21]`
10	$[-Y, -X, -Z]$	$[0, 0, \frac{1}{2}]$	`[10,-22]`
11	$[-X+Y, Y, -Z]$	$[0, 0, \frac{1}{2}]$	`[11,-23]`
12	$[X, X-Y, -Z]$	$[0, 0, \frac{1}{2}]$	`[12,-24]`

* Wyckoff site: `2c`, site symmetry: `-6m2`

Table 15: Wyckoff bond: `2a@2c`, bond symmetry: `-6m2`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	`[1,2,3,-10,-11,-12,-16,-17,-18,19,20,21]`
2	$[0, 0, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[4,5,6,-7,-8,-9,-13,-14,-15,22,23,24]`

Table 16: Wyckoff bond: `6b@2c`, bond symmetry: `mm2`

No.	vector	center	mapping
1	$[X, 2X, 0]$	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	`[1,11,16,20]`
2	$[-2X, -X, 0]$	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	`[2,10,17,19]`
3	$[X, -X, 0]$	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	`[3,12,18,21]`
4	$[-X, -2X, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[4,8,13,23]`
5	$[2X, X, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[5,7,14,22]`
6	$[-X, X, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[6,9,15,24]`

Table 17: Wyckoff bond: `6c@2c`, bond symmetry: `mm2`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	`[1,-11,16,-20]`
2	$[0, X, 0]$	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	`[2,-10,17,-19]`
3	$[-X, -X, 0]$	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	`[3,-12,18,-21]`
4	$[-X, 0, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[4,-8,13,-23]`
5	$[0, -X, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[5,-7,14,-22]`
6	$[X, X, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[6,-9,15,-24]`

Table 18: Wyckoff bond: `12d@2c`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	`[1,16]`
2	$[-Y, X-Y, 0]$	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	`[2,17]`
3	$[-X+Y, -X, 0]$	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	`[3,18]`
4	$[-X, -Y, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[4,13]`
5	$[Y, -X+Y, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[5,14]`
6	$[X-Y, X, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[6,15]`
7	$[Y, X, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[7,22]`
8	$[X-Y, -Y, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[8,23]`
9	$[-X, -X+Y, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[9,24]`
10	$[-Y, -X, 0]$	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	`[10,19]`
11	$[-X+Y, Y, 0]$	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	`[11,20]`
12	$[X, X-Y, 0]$	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	`[12,21]`

Table 19: Wyckoff bond: `12e@2c`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X, 2X, Z]$	$[\frac{1}{3}, \frac{2}{3}, 0]$	`[1,20]`
2	$[-2X, -X, Z]$	$[\frac{1}{3}, \frac{2}{3}, 0]$	`[2,19]`
3	$[X, -X, Z]$	$[\frac{1}{3}, \frac{2}{3}, 0]$	`[3,21]`
4	$[-X, -2X, Z]$	$[\frac{2}{3}, \frac{1}{3}, 0]$	`[4,23]`
5	$[2X, X, Z]$	$[\frac{2}{3}, \frac{1}{3}, 0]$	`[5,22]`
6	$[-X, X, Z]$	$[\frac{2}{3}, \frac{1}{3}, 0]$	`[6,24]`
7	$[2X, X, -Z]$	$[\frac{2}{3}, \frac{1}{3}, 0]$	`[7,14]`
8	$[-X, -2X, -Z]$	$[\frac{2}{3}, \frac{1}{3}, 0]$	`[8,13]`
9	$[-X, X, -Z]$	$[\frac{2}{3}, \frac{1}{3}, 0]$	`[9,15]`
10	$[-2X, -X, -Z]$	$[\frac{1}{3}, \frac{2}{3}, 0]$	`[10,17]`
11	$[X, 2X, -Z]$	$[\frac{1}{3}, \frac{2}{3}, 0]$	`[11,16]`
12	$[X, -X, -Z]$	$[\frac{1}{3}, \frac{2}{3}, 0]$	`[12,18]`

Table 20: Wyckoff bond: `12f@2c`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, 0, Z]$	$[\frac{1}{3}, \frac{2}{3}, 0]$	`[1,-11]`
2	$[0, X, Z]$	$[\frac{1}{3}, \frac{2}{3}, 0]$	`[2,-10]`
3	$[-X, -X, Z]$	$[\frac{1}{3}, \frac{2}{3}, 0]$	`[3,-12]`
4	$[-X, 0, Z]$	$[\frac{2}{3}, \frac{1}{3}, 0]$	`[4,-8]`
5	$[0, -X, Z]$	$[\frac{2}{3}, \frac{1}{3}, 0]$	`[5,-7]`
6	$[X, X, Z]$	$[\frac{2}{3}, \frac{1}{3}, 0]$	`[6,-9]`
7	$[-X, 0, -Z]$	$[\frac{2}{3}, \frac{1}{3}, 0]$	`[13,-23]`
8	$[0, -X, -Z]$	$[\frac{2}{3}, \frac{1}{3}, 0]$	`[14,-22]`
9	$[X, X, -Z]$	$[\frac{2}{3}, \frac{1}{3}, 0]$	`[15,-24]`
10	$[X, 0, -Z]$	$[\frac{1}{3}, \frac{2}{3}, 0]$	`[16,-20]`
11	$[0, X, -Z]$	$[\frac{1}{3}, \frac{2}{3}, 0]$	`[17,-19]`
12	$[-X, -X, -Z]$	$[\frac{1}{3}, \frac{2}{3}, 0]$	`[18,-21]`

Table 21: Wyckoff bond: `24g@2c`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[\frac{1}{3}, \frac{2}{3}, 0]$	`[1]`
2	$[-Y, X-Y, Z]$	$[\frac{1}{3}, \frac{2}{3}, 0]$	`[2]`
3	$[-X+Y, -X, Z]$	$[\frac{1}{3}, \frac{2}{3}, 0]$	`[3]`
4	$[-X, -Y, Z]$	$[\frac{2}{3}, \frac{1}{3}, 0]$	`[4]`
5	$[Y, -X+Y, Z]$	$[\frac{2}{3}, \frac{1}{3}, 0]$	`[5]`
6	$[X-Y, X, Z]$	$[\frac{2}{3}, \frac{1}{3}, 0]$	`[6]`
7	$[Y, X, -Z]$	$[\frac{2}{3}, \frac{1}{3}, 0]$	`[7]`
8	$[X-Y, -Y, -Z]$	$[\frac{2}{3}, \frac{1}{3}, 0]$	`[8]`
9	$[-X, -X+Y, -Z]$	$[\frac{2}{3}, \frac{1}{3}, 0]$	`[9]`
10	$[-Y, -X, -Z]$	$[\frac{1}{3}, \frac{2}{3}, 0]$	`[10]`
11	$[-X+Y, Y, -Z]$	$[\frac{1}{3}, \frac{2}{3}, 0]$	`[11]`
12	$[X, X-Y, -Z]$	$[\frac{1}{3}, \frac{2}{3}, 0]$	`[12]`

continued ...

Table 21

No.	vector	center	mapping
13	$[-X, -Y, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	[13]
14	$[Y, -X+Y, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	[14]
15	$[X-Y, X, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	[15]
16	$[X, Y, -Z]$	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	[16]
17	$[-Y, X-Y, -Z]$	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	[17]
18	$[-X+Y, -X, -Z]$	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	[18]
19	$[-Y, -X, Z]$	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	[19]
20	$[-X+Y, Y, Z]$	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	[20]
21	$[X, X-Y, Z]$	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	[21]
22	$[Y, X, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	[22]
23	$[X-Y, -Y, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	[23]
24	$[-X, -X+Y, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	[24]

* Wyckoff site: 2d, site symmetry: -6m2

Table 22: Wyckoff bond: 2a@2d, bond symmetry: -6m2

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	[1,2,3,-10,-11,-12,-16,-17,-18,19,20,21]
2	$[0, 0, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	[4,5,6,-7,-8,-9,-13,-14,-15,22,23,24]

Table 23: Wyckoff bond: 6b@2d, bond symmetry: mm2

No.	vector	center	mapping
1	$[X, 2X, 0]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	[1,11,16,20]
2	$[-2X, -X, 0]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	[2,10,17,19]
3	$[X, -X, 0]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	[3,12,18,21]
4	$[-X, -2X, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	[4,8,13,23]
5	$[2X, X, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	[5,7,14,22]
6	$[-X, X, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	[6,9,15,24]

Table 24: Wyckoff bond: 6c@2d, bond symmetry: mm2

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	[1,-11,16,-20]
2	$[0, X, 0]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	[2,-10,17,-19]
3	$[-X, -X, 0]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	[3,-12,18,-21]
4	$[-X, 0, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	[4,-8,13,-23]
5	$[0, -X, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	[5,-7,14,-22]
6	$[X, X, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	[6,-9,15,-24]

Table 25: Wyckoff bond: `12d@2d`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}]$	`[1,16]`
2	$[-Y, X-Y, 0]$	$[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}]$	`[2,17]`
3	$[-X+Y, -X, 0]$	$[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}]$	`[3,18]`
4	$[-X, -Y, 0]$	$[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}]$	`[4,13]`
5	$[Y, -X+Y, 0]$	$[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}]$	`[5,14]`
6	$[X-Y, X, 0]$	$[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}]$	`[6,15]`
7	$[Y, X, 0]$	$[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}]$	`[7,22]`
8	$[X-Y, -Y, 0]$	$[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}]$	`[8,23]`
9	$[-X, -X+Y, 0]$	$[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}]$	`[9,24]`
10	$[-Y, -X, 0]$	$[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}]$	`[10,19]`
11	$[-X+Y, Y, 0]$	$[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}]$	`[11,20]`
12	$[X, X-Y, 0]$	$[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}]$	`[12,21]`

Table 26: Wyckoff bond: `12e@2d`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X, 2X, Z]$	$[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}]$	`[1,20]`
2	$[-2X, -X, Z]$	$[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}]$	`[2,19]`
3	$[X, -X, Z]$	$[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}]$	`[3,21]`
4	$[-X, -2X, Z]$	$[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}]$	`[4,23]`
5	$[2X, X, Z]$	$[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}]$	`[5,22]`
6	$[-X, X, Z]$	$[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}]$	`[6,24]`
7	$[2X, X, -Z]$	$[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}]$	`[7,14]`
8	$[-X, -2X, -Z]$	$[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}]$	`[8,13]`
9	$[-X, X, -Z]$	$[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}]$	`[9,15]`
10	$[-2X, -X, -Z]$	$[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}]$	`[10,17]`
11	$[X, 2X, -Z]$	$[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}]$	`[11,16]`
12	$[X, -X, -Z]$	$[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}]$	`[12,18]`

Table 27: Wyckoff bond: `12f@2d`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, 0, Z]$	$[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}]$	`[1,-11]`
2	$[0, X, Z]$	$[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}]$	`[2,-10]`
3	$[-X, -X, Z]$	$[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}]$	`[3,-12]`
4	$[-X, 0, Z]$	$[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}]$	`[4,-8]`
5	$[0, -X, Z]$	$[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}]$	`[5,-7]`
6	$[X, X, Z]$	$[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}]$	`[6,-9]`
7	$[-X, 0, -Z]$	$[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}]$	`[13,-23]`
8	$[0, -X, -Z]$	$[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}]$	`[14,-22]`
9	$[X, X, -Z]$	$[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}]$	`[15,-24]`
10	$[X, 0, -Z]$	$[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}]$	`[16,-20]`
11	$[0, X, -Z]$	$[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}]$	`[17,-19]`
12	$[-X, -X, -Z]$	$[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}]$	`[18,-21]`

Table 28: Wyckoff bond: `24g@2d`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}]$	`[1]`
2	$[-Y, X-Y, Z]$	$[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}]$	`[2]`
3	$[-X+Y, -X, Z]$	$[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}]$	`[3]`
4	$[-X, -Y, Z]$	$[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}]$	`[4]`
5	$[Y, -X+Y, Z]$	$[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}]$	`[5]`
6	$[X-Y, X, Z]$	$[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}]$	`[6]`
7	$[Y, X, -Z]$	$[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}]$	`[7]`
8	$[X-Y, -Y, -Z]$	$[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}]$	`[8]`
9	$[-X, -X+Y, -Z]$	$[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}]$	`[9]`
10	$[-Y, -X, -Z]$	$[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}]$	`[10]`
11	$[-X+Y, Y, -Z]$	$[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}]$	`[11]`
12	$[X, X-Y, -Z]$	$[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}]$	`[12]`
13	$[-X, -Y, -Z]$	$[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}]$	`[13]`
14	$[Y, -X+Y, -Z]$	$[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}]$	`[14]`
15	$[X-Y, X, -Z]$	$[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}]$	`[15]`
16	$[X, Y, -Z]$	$[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}]$	`[16]`
17	$[-Y, X-Y, -Z]$	$[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}]$	`[17]`
18	$[-X+Y, -X, -Z]$	$[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}]$	`[18]`
19	$[-Y, -X, Z]$	$[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}]$	`[19]`
20	$[-X+Y, Y, Z]$	$[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}]$	`[20]`
21	$[X, X-Y, Z]$	$[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}]$	`[21]`
22	$[Y, X, Z]$	$[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}]$	`[22]`
23	$[X-Y, -Y, Z]$	$[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}]$	`[23]`
24	$[-X, -X+Y, Z]$	$[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}]$	`[24]`

* Wyckoff site: `2e`, site symmetry: `6mm`

Table 29: Wyckoff bond: `2a@2e`, bond symmetry: `6mm`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[0, 0, z]$	`[1,2,3,4,5,6,19,20,21,22,23,24]`
2	$[0, 0, -Z]$	$[0, 0, -z]$	`[7,8,9,10,11,12,13,14,15,16,17,18]`

Table 30: Wyckoff bond: `6b@2e`, bond symmetry: `2mm`

No.	vector	center	mapping
1	$[X, 2X, 0]$	$[0, 0, z]$	`[1,-4,20,-23]`
2	$[-2X, -X, 0]$	$[0, 0, z]$	`[2,-5,19,-22]`
3	$[X, -X, 0]$	$[0, 0, z]$	`[3,-6,21,-24]`
4	$[2X, X, 0]$	$[0, 0, -z]$	`[7,-10,14,-17]`
5	$[-X, -2X, 0]$	$[0, 0, -z]$	`[8,-11,13,-16]`
6	$[-X, X, 0]$	$[0, 0, -z]$	`[9,-12,15,-18]`

Table 31: Wyckoff bond: `6c@2e`, bond symmetry: `2mm`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[0, 0, z]$	`[1,-4,-20,23]`
2	$[0, X, 0]$	$[0, 0, z]$	`[2,-5,-19,22]`
3	$[-X, -X, 0]$	$[0, 0, z]$	`[3,-6,-21,24]`
4	$[0, X, 0]$	$[0, 0, -z]$	`[7,-10,-14,17]`
5	$[X, 0, 0]$	$[0, 0, -z]$	`[8,-11,-13,16]`
6	$[-X, -X, 0]$	$[0, 0, -z]$	`[9,-12,-15,18]`

Table 32: Wyckoff bond: `12d@2e`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[0, 0, z]$	`[1,-4]`
2	$[-Y, X-Y, 0]$	$[0, 0, z]$	`[2,-5]`
3	$[-X+Y, -X, 0]$	$[0, 0, z]$	`[3,-6]`
4	$[Y, X, 0]$	$[0, 0, -z]$	`[7,-10]`
5	$[X-Y, -Y, 0]$	$[0, 0, -z]$	`[8,-11]`
6	$[-X, -X+Y, 0]$	$[0, 0, -z]$	`[9,-12]`
7	$[-X, -Y, 0]$	$[0, 0, -z]$	`[13,-16]`
8	$[Y, -X+Y, 0]$	$[0, 0, -z]$	`[14,-17]`
9	$[X-Y, X, 0]$	$[0, 0, -z]$	`[15,-18]`
10	$[-Y, -X, 0]$	$[0, 0, z]$	`[19,-22]`
11	$[-X+Y, Y, 0]$	$[0, 0, z]$	`[20,-23]`
12	$[X, X-Y, 0]$	$[0, 0, z]$	`[21,-24]`

Table 33: Wyckoff bond: `12e@2e`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X, 2X, Z]$	$[0, 0, z]$	`[1,20]`
2	$[-2X, -X, Z]$	$[0, 0, z]$	`[2,19]`
3	$[X, -X, Z]$	$[0, 0, z]$	`[3,21]`
4	$[-X, -2X, Z]$	$[0, 0, z]$	`[4,23]`
5	$[2X, X, Z]$	$[0, 0, z]$	`[5,22]`
6	$[-X, X, Z]$	$[0, 0, z]$	`[6,24]`
7	$[2X, X, -Z]$	$[0, 0, -z]$	`[7,14]`
8	$[-X, -2X, -Z]$	$[0, 0, -z]$	`[8,13]`
9	$[-X, X, -Z]$	$[0, 0, -z]$	`[9,15]`
10	$[-2X, -X, -Z]$	$[0, 0, -z]$	`[10,17]`
11	$[X, 2X, -Z]$	$[0, 0, -z]$	`[11,16]`
12	$[X, -X, -Z]$	$[0, 0, -z]$	`[12,18]`

Table 34: Wyckoff bond: `12f@2e`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, 0, Z]$	$[0, 0, z]$	`[1,23]`

continued ...

Table 34

No.	vector	center	mapping
2	$[0, X, Z]$	$[0, 0, z]$	[2,22]
3	$[-X, -X, Z]$	$[0, 0, z]$	[3,24]
4	$[-X, 0, Z]$	$[0, 0, z]$	[4,20]
5	$[0, -X, Z]$	$[0, 0, z]$	[5,19]
6	$[X, X, Z]$	$[0, 0, z]$	[6,21]
7	$[0, X, -Z]$	$[0, 0, -z]$	[7,17]
8	$[X, 0, -Z]$	$[0, 0, -z]$	[8,16]
9	$[-X, -X, -Z]$	$[0, 0, -z]$	[9,18]
10	$[0, -X, -Z]$	$[0, 0, -z]$	[10,14]
11	$[-X, 0, -Z]$	$[0, 0, -z]$	[11,13]
12	$[X, X, -Z]$	$[0, 0, -z]$	[12,15]

Table 35: Wyckoff bond: 24g@2e, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, z]$	[1]
2	$[-Y, X-Y, Z]$	$[0, 0, z]$	[2]
3	$[-X+Y, -X, Z]$	$[0, 0, z]$	[3]
4	$[-X, -Y, Z]$	$[0, 0, z]$	[4]
5	$[Y, -X+Y, Z]$	$[0, 0, z]$	[5]
6	$[X-Y, X, Z]$	$[0, 0, z]$	[6]
7	$[Y, X, -Z]$	$[0, 0, -z]$	[7]
8	$[X-Y, -Y, -Z]$	$[0, 0, -z]$	[8]
9	$[-X, -X+Y, -Z]$	$[0, 0, -z]$	[9]
10	$[-Y, -X, -Z]$	$[0, 0, -z]$	[10]
11	$[-X+Y, Y, -Z]$	$[0, 0, -z]$	[11]
12	$[X, X-Y, -Z]$	$[0, 0, -z]$	[12]
13	$[-X, -Y, -Z]$	$[0, 0, -z]$	[13]
14	$[Y, -X+Y, -Z]$	$[0, 0, -z]$	[14]
15	$[X-Y, X, -Z]$	$[0, 0, -z]$	[15]
16	$[X, Y, -Z]$	$[0, 0, -z]$	[16]
17	$[-Y, X-Y, -Z]$	$[0, 0, -z]$	[17]
18	$[-X+Y, -X, -Z]$	$[0, 0, -z]$	[18]
19	$[-Y, -X, Z]$	$[0, 0, z]$	[19]
20	$[-X+Y, Y, Z]$	$[0, 0, z]$	[20]
21	$[X, X-Y, Z]$	$[0, 0, z]$	[21]
22	$[Y, X, Z]$	$[0, 0, z]$	[22]
23	$[X-Y, -Y, Z]$	$[0, 0, z]$	[23]
24	$[-X, -X+Y, Z]$	$[0, 0, z]$	[24]

* Wyckoff site: 3f, site symmetry: mmm

Table 36: Wyckoff bond: `3a@3f`, bond symmetry: `mmm`

No.	vector	center	mapping
1	$[X, 2X, 0]$	$\left[\frac{1}{2}, 0, 0\right]$	`[1,-4,-8,11,-13,16,20,-23]`
2	$[-2X, -X, 0]$	$\left[0, \frac{1}{2}, 0\right]$	`[2,-5,-7,10,-14,17,19,-22]`
3	$[X, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[3,-6,-9,12,-15,18,21,-24]`

Table 37: Wyckoff bond: `3b@3f`, bond symmetry: `mmm`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[\frac{1}{2}, 0, 0\right]$	`[1,-4,8,-11,-13,16,-20,23]`
2	$[0, X, 0]$	$\left[0, \frac{1}{2}, 0\right]$	`[2,-5,7,-10,-14,17,-19,22]`
3	$[-X, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[3,-6,9,-12,-15,18,-21,24]`

Table 38: Wyckoff bond: `3c@3f`, bond symmetry: `mmm`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[1,4,-8,-11,-13,-16,20,23]`
2	$[0, 0, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[2,5,-7,-10,-14,-17,19,22]`
3	$[0, 0, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[3,6,-9,-12,-15,-18,21,24]`

Table 39: Wyckoff bond: `6d@3f`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[\frac{1}{2}, 0, 0\right]$	`[1,-4,-13,16]`
2	$[-Y, X-Y, 0]$	$\left[0, \frac{1}{2}, 0\right]$	`[2,-5,-14,17]`
3	$[-X+Y, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[3,-6,-15,18]`
4	$[Y, X, 0]$	$\left[0, \frac{1}{2}, 0\right]$	`[7,-10,-19,22]`
5	$[X-Y, -Y, 0]$	$\left[\frac{1}{2}, 0, 0\right]$	`[8,-11,-20,23]`
6	$[-X, -X+Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,-12,-21,24]`

Table 40: Wyckoff bond: `6e@3f`, bond symmetry: `.2/m.`

No.	vector	center	mapping
1	$[X, 2X, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[1,-8,-13,20]`
2	$[-2X, -X, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[2,-7,-14,19]`
3	$[X, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[3,-9,-15,21]`
4	$[-X, -2X, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[4,-11,-16,23]`
5	$[2X, X, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[5,-10,-17,22]`
6	$[-X, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[6,-12,-18,24]`

Table 41: Wyckoff bond: `6f@3f`, bond symmetry: `..2/m`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[1,-11,-13,23]`
2	$[0, X, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[2,-10,-14,22]`
3	$[-X, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[3,-12,-15,24]`
4	$[-X, 0, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[4,-8,-16,20]`
5	$[0, -X, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[5,-7,-17,19]`
6	$[X, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[6,-9,-18,21]`

Table 42: Wyckoff bond: `12g@3f`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[1,-13]`
2	$[-Y, X-Y, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[2,-14]`
3	$[-X+Y, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[3,-15]`
4	$[-X, -Y, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[4,-16]`
5	$[Y, -X+Y, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[5,-17]`
6	$[X-Y, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[6,-18]`
7	$[Y, X, -Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[7,-19]`
8	$[X-Y, -Y, -Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[8,-20]`
9	$[-X, -X+Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,-21]`
10	$[-Y, -X, -Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[10,-22]`
11	$[-X+Y, Y, -Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[11,-23]`
12	$[X, X-Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[12,-24]`

* Wyckoff site: `3g`, site symmetry: `mmm`

Table 43: Wyckoff bond: `3a@3g`, bond symmetry: `mmm`

No.	vector	center	mapping
1	$[X, 2X, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[1,-4,-8,11,-13,16,20,-23]`
2	$[-2X, -X, 0]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[2,-5,-7,10,-14,17,19,-22]`
3	$[X, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[3,-6,-9,12,-15,18,21,-24]`

Table 44: Wyckoff bond: `3b@3g`, bond symmetry: `mmm`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[1,-4,8,-11,-13,16,-20,23]`
2	$[0, X, 0]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[2,-5,7,-10,-14,17,-19,22]`
3	$[-X, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[3,-6,9,-12,-15,18,-21,24]`

Table 45: Wyckoff bond: `3c@3g`, bond symmetry: `mmm`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[1,4,-8,-11,-13,-16,20,23]`

continued ...

Table 45

No.	vector	center	mapping
2	$[0,\, 0,\, Z]$	$\left[0,\, \frac{1}{2},\, \frac{1}{2}\right]$	`[2,5,-7,-10,-14,-17,19,22]`
3	$[0,\, 0,\, Z]$	$\left[\frac{1}{2},\, \frac{1}{2},\, \frac{1}{2}\right]$	`[3,6,-9,-12,-15,-18,21,24]`

Table 46: Wyckoff bond: `6d@3g`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[X,\, Y,\, 0]$	$\left[\frac{1}{2},\, 0,\, \frac{1}{2}\right]$	`[1,-4,-13,16]`
2	$[-Y,\, X-Y,\, 0]$	$\left[0,\, \frac{1}{2},\, \frac{1}{2}\right]$	`[2,-5,-14,17]`
3	$[-X+Y,\, -X,\, 0]$	$\left[\frac{1}{2},\, \frac{1}{2},\, \frac{1}{2}\right]$	`[3,-6,-15,18]`
4	$[Y,\, X,\, 0]$	$\left[0,\, \frac{1}{2},\, \frac{1}{2}\right]$	`[7,-10,-19,22]`
5	$[X-Y,\, -Y,\, 0]$	$\left[\frac{1}{2},\, 0,\, \frac{1}{2}\right]$	`[8,-11,-20,23]`
6	$[-X,\, -X+Y,\, 0]$	$\left[\frac{1}{2},\, \frac{1}{2},\, \frac{1}{2}\right]$	`[9,-12,-21,24]`

Table 47: Wyckoff bond: `6e@3g`, bond symmetry: `.2/m.`

No.	vector	center	mapping
1	$[X,\, 2X,\, Z]$	$\left[\frac{1}{2},\, 0,\, \frac{1}{2}\right]$	`[1,-8,-13,20]`
2	$[-2X,\, -X,\, Z]$	$\left[0,\, \frac{1}{2},\, \frac{1}{2}\right]$	`[2,-7,-14,19]`
3	$[X,\, -X,\, Z]$	$\left[\frac{1}{2},\, \frac{1}{2},\, \frac{1}{2}\right]$	`[3,-9,-15,21]`
4	$[-X,\, -2X,\, Z]$	$\left[\frac{1}{2},\, 0,\, \frac{1}{2}\right]$	`[4,-11,-16,23]`
5	$[2X,\, X,\, Z]$	$\left[0,\, \frac{1}{2},\, \frac{1}{2}\right]$	`[5,-10,-17,22]`
6	$[-X,\, X,\, Z]$	$\left[\frac{1}{2},\, \frac{1}{2},\, \frac{1}{2}\right]$	`[6,-12,-18,24]`

Table 48: Wyckoff bond: `6f@3g`, bond symmetry: `..2/m`

No.	vector	center	mapping
1	$[X,\, 0,\, Z]$	$\left[\frac{1}{2},\, 0,\, \frac{1}{2}\right]$	`[1,-11,-13,23]`
2	$[0,\, X,\, Z]$	$\left[0,\, \frac{1}{2},\, \frac{1}{2}\right]$	`[2,-10,-14,22]`
3	$[-X,\, -X,\, Z]$	$\left[\frac{1}{2},\, \frac{1}{2},\, \frac{1}{2}\right]$	`[3,-12,-15,24]`
4	$[-X,\, 0,\, Z]$	$\left[\frac{1}{2},\, 0,\, \frac{1}{2}\right]$	`[4,-8,-16,20]`
5	$[0,\, -X,\, Z]$	$\left[0,\, \frac{1}{2},\, \frac{1}{2}\right]$	`[5,-7,-17,19]`
6	$[X,\, X,\, Z]$	$\left[\frac{1}{2},\, \frac{1}{2},\, \frac{1}{2}\right]$	`[6,-9,-18,21]`

Table 49: Wyckoff bond: `12g@3g`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X,\, Y,\, Z]$	$\left[\frac{1}{2},\, 0,\, \frac{1}{2}\right]$	`[1,-13]`
2	$[-Y,\, X-Y,\, Z]$	$\left[0,\, \frac{1}{2},\, \frac{1}{2}\right]$	`[2,-14]`
3	$[-X+Y,\, -X,\, Z]$	$\left[\frac{1}{2},\, \frac{1}{2},\, \frac{1}{2}\right]$	`[3,-15]`
4	$[-X,\, -Y,\, Z]$	$\left[\frac{1}{2},\, 0,\, \frac{1}{2}\right]$	`[4,-16]`
5	$[Y,\, -X+Y,\, Z]$	$\left[0,\, \frac{1}{2},\, \frac{1}{2}\right]$	`[5,-17]`
6	$[X-Y,\, X,\, Z]$	$\left[\frac{1}{2},\, \frac{1}{2},\, \frac{1}{2}\right]$	`[6,-18]`

continued ...

Table 49

No.	vector	center	mapping
7	$[Y, X, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[7,-19]`
8	$[X-Y, -Y, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[8,-20]`
9	$[-X, -X+Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[9,-21]`
10	$[-Y, -X, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[10,-22]`
11	$[-X+Y, Y, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[11,-23]`
12	$[X, X-Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[12,-24]`

* Wyckoff site: `4h`, site symmetry: `3m`.

Table 50: Wyckoff bond: `4a@4h`, bond symmetry: `3m`.

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	`[1,2,3,19,20,21]`
2	$[0, 0, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, z\right]$	`[4,5,6,22,23,24]`
3	$[0, 0, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	`[7,8,9,13,14,15]`
4	$[0, 0, -Z]$	$\left[\frac{1}{3}, \frac{2}{3}, -z\right]$	`[10,11,12,16,17,18]`

Table 51: Wyckoff bond: `12b@4h`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X, 2X, Z]$	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	`[1,20]`
2	$[-2X, -X, Z]$	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	`[2,19]`
3	$[X, -X, Z]$	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	`[3,21]`
4	$[-X, -2X, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, z\right]$	`[4,23]`
5	$[2X, X, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, z\right]$	`[5,22]`
6	$[-X, X, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, z\right]$	`[6,24]`
7	$[2X, X, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	`[7,14]`
8	$[-X, -2X, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	`[8,13]`
9	$[-X, X, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	`[9,15]`
10	$[-2X, -X, -Z]$	$\left[\frac{1}{3}, \frac{2}{3}, -z\right]$	`[10,17]`
11	$[X, 2X, -Z]$	$\left[\frac{1}{3}, \frac{2}{3}, -z\right]$	`[11,16]`
12	$[X, -X, -Z]$	$\left[\frac{1}{3}, \frac{2}{3}, -z\right]$	`[12,18]`

Table 52: Wyckoff bond: `12c@4h`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	`[1,-20]`
2	$[0, X, 0]$	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	`[2,-19]`
3	$[-X, -X, 0]$	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	`[3,-21]`
4	$[-X, 0, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, z\right]$	`[4,-23]`
5	$[0, -X, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, z\right]$	`[5,-22]`
6	$[X, X, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, z\right]$	`[6,-24]`
7	$[0, X, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	`[7,-14]`
8	$[X, 0, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	`[8,-13]`

continued ...

Table 52

No.	vector	center	mapping
9	$[-X, -X, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	`[9,-15]`
10	$[0, -X, 0]$	$\left[\frac{1}{3}, \frac{2}{3}, -z\right]$	`[10,-17]`
11	$[-X, 0, 0]$	$\left[\frac{1}{3}, \frac{2}{3}, -z\right]$	`[11,-16]`
12	$[X, X, 0]$	$\left[\frac{1}{3}, \frac{2}{3}, -z\right]$	`[12,-18]`

Table 53: Wyckoff bond: `24d@4h`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	`[1]`
2	$[-Y, X-Y, Z]$	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	`[2]`
3	$[-X+Y, -X, Z]$	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	`[3]`
4	$[-X, -Y, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, z\right]$	`[4]`
5	$[Y, -X+Y, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, z\right]$	`[5]`
6	$[X-Y, X, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, z\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	`[7]`
8	$[X-Y, -Y, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	`[8]`
9	$[-X, -X+Y, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	`[9]`
10	$[-Y, -X, -Z]$	$\left[\frac{1}{3}, \frac{2}{3}, -z\right]$	`[10]`
11	$[-X+Y, Y, -Z]$	$\left[\frac{1}{3}, \frac{2}{3}, -z\right]$	`[11]`
12	$[X, X-Y, -Z]$	$\left[\frac{1}{3}, \frac{2}{3}, -z\right]$	`[12]`
13	$[-X, -Y, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	`[13]`
14	$[Y, -X+Y, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	`[14]`
15	$[X-Y, X, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	`[15]`
16	$[X, Y, -Z]$	$\left[\frac{1}{3}, \frac{2}{3}, -z\right]$	`[16]`
17	$[-Y, X-Y, -Z]$	$\left[\frac{1}{3}, \frac{2}{3}, -z\right]$	`[17]`
18	$[-X+Y, -X, -Z]$	$\left[\frac{1}{3}, \frac{2}{3}, -z\right]$	`[18]`
19	$[-Y, -X, Z]$	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	`[19]`
20	$[-X+Y, Y, Z]$	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	`[20]`
21	$[X, X-Y, Z]$	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	`[21]`
22	$[Y, X, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, z\right]$	`[22]`
23	$[X-Y, -Y, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, z\right]$	`[23]`
24	$[-X, -X+Y, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, z\right]$	`[24]`

* Wyckoff site: `6i`, site symmetry: `2mm`

Table 54: Wyckoff bond: `6a@6i`, bond symmetry: `2mm`

No.	vector	center	mapping
1	$[X, 2X, 0]$	$\left[\frac{1}{2}, 0, z\right]$	`[1,-4,20,-23]`
2	$[-2X, -X, 0]$	$\left[0, \frac{1}{2}, z\right]$	`[2,-5,19,-22]`
3	$[X, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[3,-6,21,-24]`
4	$[2X, X, 0]$	$\left[0, \frac{1}{2}, -z\right]$	`[7,-10,14,-17]`
5	$[-X, -2X, 0]$	$\left[\frac{1}{2}, 0, -z\right]$	`[8,-11,13,-16]`
6	$[-X, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[9,-12,15,-18]`

Table 55: Wyckoff bond: `6b@6i`, bond symmetry: `2mm`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[\frac{1}{2}, 0, z\right]$	`[1,-4,-20,23]`
2	$[0, X, 0]$	$\left[0, \frac{1}{2}, z\right]$	`[2,-5,-19,22]`
3	$[-X, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[3,-6,-21,24]`
4	$[0, X, 0]$	$\left[0, \frac{1}{2}, -z\right]$	`[7,-10,-14,17]`
5	$[X, 0, 0]$	$\left[\frac{1}{2}, 0, -z\right]$	`[8,-11,-13,16]`
6	$[-X, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[9,-12,-15,18]`

Table 56: Wyckoff bond: `6c@6i`, bond symmetry: `2mm`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[\frac{1}{2}, 0, z\right]$	`[1,4,20,23]`
2	$[0, 0, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[2,5,19,22]`
3	$[0, 0, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[3,6,21,24]`
4	$[0, 0, -Z]$	$\left[0, \frac{1}{2}, -z\right]$	`[7,10,14,17]`
5	$[0, 0, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	`[8,11,13,16]`
6	$[0, 0, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[9,12,15,18]`

Table 57: Wyckoff bond: `12d@6i`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[\frac{1}{2}, 0, z\right]$	`[1,-4]`
2	$[-Y, X-Y, 0]$	$\left[0, \frac{1}{2}, z\right]$	`[2,-5]`
3	$[-X+Y, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[3,-6]`
4	$[Y, X, 0]$	$\left[0, \frac{1}{2}, -z\right]$	`[7,-10]`
5	$[X-Y, -Y, 0]$	$\left[\frac{1}{2}, 0, -z\right]$	`[8,-11]`
6	$[-X, -X+Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[9,-12]`
7	$[-X, -Y, 0]$	$\left[\frac{1}{2}, 0, -z\right]$	`[13,-16]`
8	$[Y, -X+Y, 0]$	$\left[0, \frac{1}{2}, -z\right]$	`[14,-17]`
9	$[X-Y, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[15,-18]`
10	$[-Y, -X, 0]$	$\left[0, \frac{1}{2}, z\right]$	`[19,-22]`
11	$[-X+Y, Y, 0]$	$\left[\frac{1}{2}, 0, z\right]$	`[20,-23]`
12	$[X, X-Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[21,-24]`

Table 58: Wyckoff bond: `12e@6i`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X, 2X, Z]$	$\left[\frac{1}{2}, 0, z\right]$	`[1,20]`
2	$[-2X, -X, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[2,19]`
3	$[X, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[3,21]`
4	$[-X, -2X, Z]$	$\left[\frac{1}{2}, 0, z\right]$	`[4,23]`
5	$[2X, X, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[5,22]`
6	$[-X, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[6,24]`
7	$[2X, X, -Z]$	$\left[0, \frac{1}{2}, -z\right]$	`[7,14]`

continued ...

Table 58

No.	vector	center	mapping
8	$[-X, -2X, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	[8,13]
9	$[-X, X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	[9,15]
10	$[-2X, -X, -Z]$	$\left[0, \frac{1}{2}, -z\right]$	[10,17]
11	$[X, 2X, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	[11,16]
12	$[X, -X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	[12,18]

Table 59: Wyckoff bond: 12f@6i, bond symmetry: ..m

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[\frac{1}{2}, 0, z\right]$	[1,23]
2	$[0, X, Z]$	$\left[0, \frac{1}{2}, z\right]$	[2,22]
3	$[-X, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	[3,24]
4	$[-X, 0, Z]$	$\left[\frac{1}{2}, 0, z\right]$	[4,20]
5	$[0, -X, Z]$	$\left[0, \frac{1}{2}, z\right]$	[5,19]
6	$[X, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	[6,21]
7	$[0, X, -Z]$	$\left[0, \frac{1}{2}, -z\right]$	[7,17]
8	$[X, 0, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	[8,16]
9	$[-X, -X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	[9,18]
10	$[0, -X, -Z]$	$\left[0, \frac{1}{2}, -z\right]$	[10,14]
11	$[-X, 0, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	[11,13]
12	$[X, X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	[12,15]

Table 60: Wyckoff bond: 24g@6i, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{2}, 0, z\right]$	[1]
2	$[-Y, X-Y, Z]$	$\left[0, \frac{1}{2}, z\right]$	[2]
3	$[-X+Y, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	[3]
4	$[-X, -Y, Z]$	$\left[\frac{1}{2}, 0, z\right]$	[4]
5	$[Y, -X+Y, Z]$	$\left[0, \frac{1}{2}, z\right]$	[5]
6	$[X-Y, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	[6]
7	$[Y, X, -Z]$	$\left[0, \frac{1}{2}, -z\right]$	[7]
8	$[X-Y, -Y, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	[8]
9	$[-X, -X+Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	[9]
10	$[-Y, -X, -Z]$	$\left[0, \frac{1}{2}, -z\right]$	[10]
11	$[-X+Y, Y, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	[11]
12	$[X, X-Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	[12]
13	$[-X, -Y, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	[13]
14	$[Y, -X+Y, -Z]$	$\left[0, \frac{1}{2}, -z\right]$	[14]
15	$[X-Y, X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	[15]
16	$[X, Y, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	[16]
17	$[-Y, X-Y, -Z]$	$\left[0, \frac{1}{2}, -z\right]$	[17]
18	$[-X+Y, -X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	[18]
19	$[-Y, -X, Z]$	$\left[0, \frac{1}{2}, z\right]$	[19]

continued ...

Table 60

No.	vector	center	mapping
20	$[-X+Y,\, Y,\, Z]$	$\left[\frac{1}{2},\, 0,\, z\right]$	`[20]`
21	$[X,\, X-Y,\, Z]$	$\left[\frac{1}{2},\, \frac{1}{2},\, z\right]$	`[21]`
22	$[Y,\, X,\, Z]$	$\left[0,\, \frac{1}{2},\, z\right]$	`[22]`
23	$[X-Y,\, -Y,\, Z]$	$\left[\frac{1}{2},\, 0,\, z\right]$	`[23]`
24	$[-X,\, -X+Y,\, Z]$	$\left[\frac{1}{2},\, \frac{1}{2},\, z\right]$	`[24]`

* Wyckoff site: `6j`, site symmetry: `m2m`

Table 61: Wyckoff bond: `6a@6j`, bond symmetry: `m2m`

No.	vector	center	mapping
1	$[X,\, 2X,\, 0]$	$[x,\, 0,\, 0]$	`[1,-8,16,-23]`
2	$[-2X,\, -X,\, 0]$	$[0,\, x,\, 0]$	`[2,-7,17,-22]`
3	$[X,\, -X,\, 0]$	$[-x,\, -x,\, 0]$	`[3,-9,18,-24]`
4	$[-X,\, -2X,\, 0]$	$[-x,\, 0,\, 0]$	`[4,-11,13,-20]`
5	$[2X,\, X,\, 0]$	$[0,\, -x,\, 0]$	`[5,-10,14,-19]`
6	$[-X,\, X,\, 0]$	$[x,\, x,\, 0]$	`[6,-12,15,-21]`

Table 62: Wyckoff bond: `6b@6j`, bond symmetry: `m2m`

No.	vector	center	mapping
1	$[X,\, 0,\, 0]$	$[x,\, 0,\, 0]$	`[1,8,16,23]`
2	$[0,\, X,\, 0]$	$[0,\, x,\, 0]$	`[2,7,17,22]`
3	$[-X,\, -X,\, 0]$	$[-x,\, -x,\, 0]$	`[3,9,18,24]`
4	$[-X,\, 0,\, 0]$	$[-x,\, 0,\, 0]$	`[4,11,13,20]`
5	$[0,\, -X,\, 0]$	$[0,\, -x,\, 0]$	`[5,10,14,19]`
6	$[X,\, X,\, 0]$	$[x,\, x,\, 0]$	`[6,12,15,21]`

Table 63: Wyckoff bond: `6c@6j`, bond symmetry: `m2m`

No.	vector	center	mapping
1	$[0,\, 0,\, Z]$	$[x,\, 0,\, 0]$	`[1,-8,-16,23]`
2	$[0,\, 0,\, Z]$	$[0,\, x,\, 0]$	`[2,-7,-17,22]`
3	$[0,\, 0,\, Z]$	$[-x,\, -x,\, 0]$	`[3,-9,-18,24]`
4	$[0,\, 0,\, Z]$	$[-x,\, 0,\, 0]$	`[4,-11,-13,20]`
5	$[0,\, 0,\, Z]$	$[0,\, -x,\, 0]$	`[5,-10,-14,19]`
6	$[0,\, 0,\, Z]$	$[x,\, x,\, 0]$	`[6,-12,-15,21]`

Table 64: Wyckoff bond: `12d@6j`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X,\, Y,\, 0]$	$[x,\, 0,\, 0]$	`[1,16]`
2	$[-Y,\, X-Y,\, 0]$	$[0,\, x,\, 0]$	`[2,17]`

continued ...

Table 64

No.	vector	center	mapping
3	$[-X+Y, -X, 0]$	$[-x, -x, 0]$	`[3,18]`
4	$[-X, -Y, 0]$	$[-x, 0, 0]$	`[4,13]`
5	$[Y, -X+Y, 0]$	$[0, -x, 0]$	`[5,14]`
6	$[X-Y, X, 0]$	$[x, x, 0]$	`[6,15]`
7	$[Y, X, 0]$	$[0, x, 0]$	`[7,22]`
8	$[X-Y, -Y, 0]$	$[x, 0, 0]$	`[8,23]`
9	$[-X, -X+Y, 0]$	$[-x, -x, 0]$	`[9,24]`
10	$[-Y, -X, 0]$	$[0, -x, 0]$	`[10,19]`
11	$[-X+Y, Y, 0]$	$[-x, 0, 0]$	`[11,20]`
12	$[X, X-Y, 0]$	$[x, x, 0]$	`[12,21]`

Table 65: Wyckoff bond: `12e@6j`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 2X, Z]$	$[x, 0, 0]$	`[1,-8]`
2	$[-2X, -X, Z]$	$[0, x, 0]$	`[2,-7]`
3	$[X, -X, Z]$	$[-x, -x, 0]$	`[3,-9]`
4	$[-X, -2X, Z]$	$[-x, 0, 0]$	`[4,-11]`
5	$[2X, X, Z]$	$[0, -x, 0]$	`[5,-10]`
6	$[-X, X, Z]$	$[x, x, 0]$	`[6,-12]`
7	$[-X, -2X, -Z]$	$[-x, 0, 0]$	`[13,-20]`
8	$[2X, X, -Z]$	$[0, -x, 0]$	`[14,-19]`
9	$[-X, X, -Z]$	$[x, x, 0]$	`[15,-21]`
10	$[X, 2X, -Z]$	$[x, 0, 0]$	`[16,-23]`
11	$[-2X, -X, -Z]$	$[0, x, 0]$	`[17,-22]`
12	$[X, -X, -Z]$	$[-x, -x, 0]$	`[18,-24]`

Table 66: Wyckoff bond: `12f@6j`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, 0, Z]$	$[x, 0, 0]$	`[1,23]`
2	$[0, X, Z]$	$[0, x, 0]$	`[2,22]`
3	$[-X, -X, Z]$	$[-x, -x, 0]$	`[3,24]`
4	$[-X, 0, Z]$	$[-x, 0, 0]$	`[4,20]`
5	$[0, -X, Z]$	$[0, -x, 0]$	`[5,19]`
6	$[X, X, Z]$	$[x, x, 0]$	`[6,21]`
7	$[0, X, -Z]$	$[0, x, 0]$	`[7,17]`
8	$[X, 0, -Z]$	$[x, 0, 0]$	`[8,16]`
9	$[-X, -X, -Z]$	$[-x, -x, 0]$	`[9,18]`
10	$[0, -X, -Z]$	$[0, -x, 0]$	`[10,14]`
11	$[-X, 0, -Z]$	$[-x, 0, 0]$	`[11,13]`
12	$[X, X, -Z]$	$[x, x, 0]$	`[12,15]`

Table 67: Wyckoff bond: `24g@6j`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, 0, 0]$	`[1]`
2	$[-Y, X-Y, Z]$	$[0, x, 0]$	`[2]`
3	$[-X+Y, -X, Z]$	$[-x, -x, 0]$	`[3]`
4	$[-X, -Y, Z]$	$[-x, 0, 0]$	`[4]`
5	$[Y, -X+Y, Z]$	$[0, -x, 0]$	`[5]`
6	$[X-Y, X, Z]$	$[x, x, 0]$	`[6]`
7	$[Y, X, -Z]$	$[0, x, 0]$	`[7]`
8	$[X-Y, -Y, -Z]$	$[x, 0, 0]$	`[8]`
9	$[-X, -X+Y, -Z]$	$[-x, -x, 0]$	`[9]`
10	$[-Y, -X, -Z]$	$[0, -x, 0]$	`[10]`
11	$[-X+Y, Y, -Z]$	$[-x, 0, 0]$	`[11]`
12	$[X, X-Y, -Z]$	$[x, x, 0]$	`[12]`
13	$[-X, -Y, -Z]$	$[-x, 0, 0]$	`[13]`
14	$[Y, -X+Y, -Z]$	$[0, -x, 0]$	`[14]`
15	$[X-Y, X, -Z]$	$[x, x, 0]$	`[15]`
16	$[X, Y, -Z]$	$[x, 0, 0]$	`[16]`
17	$[-Y, X-Y, -Z]$	$[0, x, 0]$	`[17]`
18	$[-X+Y, -X, -Z]$	$[-x, -x, 0]$	`[18]`
19	$[-Y, -X, Z]$	$[0, -x, 0]$	`[19]`
20	$[-X+Y, Y, Z]$	$[-x, 0, 0]$	`[20]`
21	$[X, X-Y, Z]$	$[x, x, 0]$	`[21]`
22	$[Y, X, Z]$	$[0, x, 0]$	`[22]`
23	$[X-Y, -Y, Z]$	$[x, 0, 0]$	`[23]`
24	$[-X, -X+Y, Z]$	$[-x, -x, 0]$	`[24]`

* Wyckoff site: `6k`, site symmetry: `m2m`

Table 68: Wyckoff bond: `6a@6k`, bond symmetry: `m2m`

No.	vector	center	mapping
1	$[X, 2X, 0]$	$\left[x, 0, \frac{1}{2}\right]$	`[1,-8,16,-23]`
2	$[-2X, -X, 0]$	$\left[0, x, \frac{1}{2}\right]$	`[2,-7,17,-22]`
3	$[X, -X, 0]$	$\left[-x, -x, \frac{1}{2}\right]$	`[3,-9,18,-24]`
4	$[-X, -2X, 0]$	$\left[-x, 0, \frac{1}{2}\right]$	`[4,-11,13,-20]`
5	$[2X, X, 0]$	$\left[0, -x, \frac{1}{2}\right]$	`[5,-10,14,-19]`
6	$[-X, X, 0]$	$\left[x, x, \frac{1}{2}\right]$	`[6,-12,15,-21]`

Table 69: Wyckoff bond: `6b@6k`, bond symmetry: `m2m`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[x, 0, \frac{1}{2}\right]$	`[1,8,16,23]`
2	$[0, X, 0]$	$\left[0, x, \frac{1}{2}\right]$	`[2,7,17,22]`
3	$[-X, -X, 0]$	$\left[-x, -x, \frac{1}{2}\right]$	`[3,9,18,24]`
4	$[-X, 0, 0]$	$\left[-x, 0, \frac{1}{2}\right]$	`[4,11,13,20]`
5	$[0, -X, 0]$	$\left[0, -x, \frac{1}{2}\right]$	`[5,10,14,19]`

continued ...

Table 69

No.	vector	center	mapping
6	$[X, X, 0]$	$\left[x, x, \frac{1}{2}\right]$	`[6,12,15,21]`

Table 70: Wyckoff bond: `6c@6k`, bond symmetry: `m2m`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[x, 0, \frac{1}{2}\right]$	`[1,-8,-16,23]`
2	$[0, 0, Z]$	$\left[0, x, \frac{1}{2}\right]$	`[2,-7,-17,22]`
3	$[0, 0, Z]$	$\left[-x, -x, \frac{1}{2}\right]$	`[3,-9,-18,24]`
4	$[0, 0, Z]$	$\left[-x, 0, \frac{1}{2}\right]$	`[4,-11,-13,20]`
5	$[0, 0, Z]$	$\left[0, -x, \frac{1}{2}\right]$	`[5,-10,-14,19]`
6	$[0, 0, Z]$	$\left[x, x, \frac{1}{2}\right]$	`[6,-12,-15,21]`

Table 71: Wyckoff bond: `12d@6k`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[x, 0, \frac{1}{2}\right]$	`[1,16]`
2	$[-Y, X-Y, 0]$	$\left[0, x, \frac{1}{2}\right]$	`[2,17]`
3	$[-X+Y, -X, 0]$	$\left[-x, -x, \frac{1}{2}\right]$	`[3,18]`
4	$[-X, -Y, 0]$	$\left[-x, 0, \frac{1}{2}\right]$	`[4,13]`
5	$[Y, -X+Y, 0]$	$\left[0, -x, \frac{1}{2}\right]$	`[5,14]`
6	$[X-Y, X, 0]$	$\left[x, x, \frac{1}{2}\right]$	`[6,15]`
7	$[Y, X, 0]$	$\left[0, x, \frac{1}{2}\right]$	`[7,22]`
8	$[X-Y, -Y, 0]$	$\left[x, 0, \frac{1}{2}\right]$	`[8,23]`
9	$[-X, -X+Y, 0]$	$\left[-x, -x, \frac{1}{2}\right]$	`[9,24]`
10	$[-Y, -X, 0]$	$\left[0, -x, \frac{1}{2}\right]$	`[10,19]`
11	$[-X+Y, Y, 0]$	$\left[-x, 0, \frac{1}{2}\right]$	`[11,20]`
12	$[X, X-Y, 0]$	$\left[x, x, \frac{1}{2}\right]$	`[12,21]`

Table 72: Wyckoff bond: `12e@6k`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 2X, Z]$	$\left[x, 0, \frac{1}{2}\right]$	`[1,-8]`
2	$[-2X, -X, Z]$	$\left[0, x, \frac{1}{2}\right]$	`[2,-7]`
3	$[X, -X, Z]$	$\left[-x, -x, \frac{1}{2}\right]$	`[3,-9]`
4	$[-X, -2X, Z]$	$\left[-x, 0, \frac{1}{2}\right]$	`[4,-11]`
5	$[2X, X, Z]$	$\left[0, -x, \frac{1}{2}\right]$	`[5,-10]`
6	$[-X, X, Z]$	$\left[x, x, \frac{1}{2}\right]$	`[6,-12]`
7	$[-X, -2X, -Z]$	$\left[-x, 0, \frac{1}{2}\right]$	`[13,-20]`
8	$[2X, X, -Z]$	$\left[0, -x, \frac{1}{2}\right]$	`[14,-19]`
9	$[-X, X, -Z]$	$\left[x, x, \frac{1}{2}\right]$	`[15,-21]`
10	$[X, 2X, -Z]$	$\left[x, 0, \frac{1}{2}\right]$	`[16,-23]`
11	$[-2X, -X, -Z]$	$\left[0, x, \frac{1}{2}\right]$	`[17,-22]`
12	$[X, -X, -Z]$	$\left[-x, -x, \frac{1}{2}\right]$	`[18,-24]`

Table 73: Wyckoff bond: `12f@6k`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[x, 0, \frac{1}{2}\right]$	`[1,23]`
2	$[0, X, Z]$	$\left[0, x, \frac{1}{2}\right]$	`[2,22]`
3	$[-X, -X, Z]$	$\left[-x, -x, \frac{1}{2}\right]$	`[3,24]`
4	$[-X, 0, Z]$	$\left[-x, 0, \frac{1}{2}\right]$	`[4,20]`
5	$[0, -X, Z]$	$\left[0, -x, \frac{1}{2}\right]$	`[5,19]`
6	$[X, X, Z]$	$\left[x, x, \frac{1}{2}\right]$	`[6,21]`
7	$[0, X, -Z]$	$\left[0, x, \frac{1}{2}\right]$	`[7,17]`
8	$[X, 0, -Z]$	$\left[x, 0, \frac{1}{2}\right]$	`[8,16]`
9	$[-X, -X, -Z]$	$\left[-x, -x, \frac{1}{2}\right]$	`[9,18]`
10	$[0, -X, -Z]$	$\left[0, -x, \frac{1}{2}\right]$	`[10,14]`
11	$[-X, 0, -Z]$	$\left[-x, 0, \frac{1}{2}\right]$	`[11,13]`
12	$[X, X, -Z]$	$\left[x, x, \frac{1}{2}\right]$	`[12,15]`

Table 74: Wyckoff bond: `24g@6k`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, 0, \frac{1}{2}\right]$	`[1]`
2	$[-Y, X-Y, Z]$	$\left[0, x, \frac{1}{2}\right]$	`[2]`
3	$[-X+Y, -X, Z]$	$\left[-x, -x, \frac{1}{2}\right]$	`[3]`
4	$[-X, -Y, Z]$	$\left[-x, 0, \frac{1}{2}\right]$	`[4]`
5	$[Y, -X+Y, Z]$	$\left[0, -x, \frac{1}{2}\right]$	`[5]`
6	$[X-Y, X, Z]$	$\left[x, x, \frac{1}{2}\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[0, x, \frac{1}{2}\right]$	`[7]`
8	$[X-Y, -Y, -Z]$	$\left[x, 0, \frac{1}{2}\right]$	`[8]`
9	$[-X, -X+Y, -Z]$	$\left[-x, -x, \frac{1}{2}\right]$	`[9]`
10	$[-Y, -X, -Z]$	$\left[0, -x, \frac{1}{2}\right]$	`[10]`
11	$[-X+Y, Y, -Z]$	$\left[-x, 0, \frac{1}{2}\right]$	`[11]`
12	$[X, X-Y, -Z]$	$\left[x, x, \frac{1}{2}\right]$	`[12]`
13	$[-X, -Y, -Z]$	$\left[-x, 0, \frac{1}{2}\right]$	`[13]`
14	$[Y, -X+Y, -Z]$	$\left[0, -x, \frac{1}{2}\right]$	`[14]`
15	$[X-Y, X, -Z]$	$\left[x, x, \frac{1}{2}\right]$	`[15]`
16	$[X, Y, -Z]$	$\left[x, 0, \frac{1}{2}\right]$	`[16]`
17	$[-Y, X-Y, -Z]$	$\left[0, x, \frac{1}{2}\right]$	`[17]`
18	$[-X+Y, -X, -Z]$	$\left[-x, -x, \frac{1}{2}\right]$	`[18]`
19	$[-Y, -X, Z]$	$\left[0, -x, \frac{1}{2}\right]$	`[19]`
20	$[-X+Y, Y, Z]$	$\left[-x, 0, \frac{1}{2}\right]$	`[20]`
21	$[X, X-Y, Z]$	$\left[x, x, \frac{1}{2}\right]$	`[21]`
22	$[Y, X, Z]$	$\left[0, x, \frac{1}{2}\right]$	`[22]`
23	$[X-Y, -Y, Z]$	$\left[x, 0, \frac{1}{2}\right]$	`[23]`
24	$[-X, -X+Y, Z]$	$\left[-x, -x, \frac{1}{2}\right]$	`[24]`

* Wyckoff site: `6l`, site symmetry: `mm2`

Table 75: Wyckoff bond: `6a@6l`, bond symmetry: `mm2`

No.	vector	center	mapping
1	$[X, 2X, 0]$	$[x, 2x, 0]$	`[1,11,16,20]`
2	$[-2X, -X, 0]$	$[-2x, -x, 0]$	`[2,10,17,19]`
3	$[X, -X, 0]$	$[x, -x, 0]$	`[3,12,18,21]`
4	$[-X, -2X, 0]$	$[-x, -2x, 0]$	`[4,8,13,23]`
5	$[2X, X, 0]$	$[2x, x, 0]$	`[5,7,14,22]`
6	$[-X, X, 0]$	$[-x, x, 0]$	`[6,9,15,24]`

Table 76: Wyckoff bond: `6b@6l`, bond symmetry: `mm2`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[x, 2x, 0]$	`[1,-11,16,-20]`
2	$[0, X, 0]$	$[-2x, -x, 0]$	`[2,-10,17,-19]`
3	$[-X, -X, 0]$	$[x, -x, 0]$	`[3,-12,18,-21]`
4	$[-X, 0, 0]$	$[-x, -2x, 0]$	`[4,-8,13,-23]`
5	$[0, -X, 0]$	$[2x, x, 0]$	`[5,-7,14,-22]`
6	$[X, X, 0]$	$[-x, x, 0]$	`[6,-9,15,-24]`

Table 77: Wyckoff bond: `6c@6l`, bond symmetry: `mm2`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[x, 2x, 0]$	`[1,-11,-16,20]`
2	$[0, 0, Z]$	$[-2x, -x, 0]$	`[2,-10,-17,19]`
3	$[0, 0, Z]$	$[x, -x, 0]$	`[3,-12,-18,21]`
4	$[0, 0, Z]$	$[-x, -2x, 0]$	`[4,-8,-13,23]`
5	$[0, 0, Z]$	$[2x, x, 0]$	`[5,-7,-14,22]`
6	$[0, 0, Z]$	$[-x, x, 0]$	`[6,-9,-15,24]`

Table 78: Wyckoff bond: `12d@6l`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[x, 2x, 0]$	`[1,16]`
2	$[-Y, X-Y, 0]$	$[-2x, -x, 0]$	`[2,17]`
3	$[-X+Y, -X, 0]$	$[x, -x, 0]$	`[3,18]`
4	$[-X, -Y, 0]$	$[-x, -2x, 0]$	`[4,13]`
5	$[Y, -X+Y, 0]$	$[2x, x, 0]$	`[5,14]`
6	$[X-Y, X, 0]$	$[-x, x, 0]$	`[6,15]`
7	$[Y, X, 0]$	$[2x, x, 0]$	`[7,22]`
8	$[X-Y, -Y, 0]$	$[-x, -2x, 0]$	`[8,23]`
9	$[-X, -X+Y, 0]$	$[-x, x, 0]$	`[9,24]`
10	$[-Y, -X, 0]$	$[-2x, -x, 0]$	`[10,19]`
11	$[-X+Y, Y, 0]$	$[x, 2x, 0]$	`[11,20]`
12	$[X, X-Y, 0]$	$[x, -x, 0]$	`[12,21]`

Table 79: Wyckoff bond: `12e@61`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X, 2X, Z]$	$[x, 2x, 0]$	`[1,20]`
2	$[-2X, -X, Z]$	$[-2x, -x, 0]$	`[2,19]`
3	$[X, -X, Z]$	$[x, -x, 0]$	`[3,21]`
4	$[-X, -2X, Z]$	$[-x, -2x, 0]$	`[4,23]`
5	$[2X, X, Z]$	$[2x, x, 0]$	`[5,22]`
6	$[-X, X, Z]$	$[-x, x, 0]$	`[6,24]`
7	$[2X, X, -Z]$	$[2x, x, 0]$	`[7,14]`
8	$[-X, -2X, -Z]$	$[-x, -2x, 0]$	`[8,13]`
9	$[-X, X, -Z]$	$[-x, x, 0]$	`[9,15]`
10	$[-2X, -X, -Z]$	$[-2x, -x, 0]$	`[10,17]`
11	$[X, 2X, -Z]$	$[x, 2x, 0]$	`[11,16]`
12	$[X, -X, -Z]$	$[x, -x, 0]$	`[12,18]`

Table 80: Wyckoff bond: `12f@61`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, 0, Z]$	$[x, 2x, 0]$	`[1,-11]`
2	$[0, X, Z]$	$[-2x, -x, 0]$	`[2,-10]`
3	$[-X, -X, Z]$	$[x, -x, 0]$	`[3,-12]`
4	$[-X, 0, Z]$	$[-x, -2x, 0]$	`[4,-8]`
5	$[0, -X, Z]$	$[2x, x, 0]$	`[5,-7]`
6	$[X, X, Z]$	$[-x, x, 0]$	`[6,-9]`
7	$[-X, 0, -Z]$	$[-x, -2x, 0]$	`[13,-23]`
8	$[0, -X, -Z]$	$[2x, x, 0]$	`[14,-22]`
9	$[X, X, -Z]$	$[-x, x, 0]$	`[15,-24]`
10	$[X, 0, -Z]$	$[x, 2x, 0]$	`[16,-20]`
11	$[0, X, -Z]$	$[-2x, -x, 0]$	`[17,-19]`
12	$[-X, -X, -Z]$	$[x, -x, 0]$	`[18,-21]`

Table 81: Wyckoff bond: `24g@61`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, 2x, 0]$	`[1]`
2	$[-Y, X-Y, Z]$	$[-2x, -x, 0]$	`[2]`
3	$[-X+Y, -X, Z]$	$[x, -x, 0]$	`[3]`
4	$[-X, -Y, Z]$	$[-x, -2x, 0]$	`[4]`
5	$[Y, -X+Y, Z]$	$[2x, x, 0]$	`[5]`
6	$[X-Y, X, Z]$	$[-x, x, 0]$	`[6]`
7	$[Y, X, -Z]$	$[2x, x, 0]$	`[7]`
8	$[X-Y, -Y, -Z]$	$[-x, -2x, 0]$	`[8]`
9	$[-X, -X+Y, -Z]$	$[-x, x, 0]$	`[9]`
10	$[-Y, -X, -Z]$	$[-2x, -x, 0]$	`[10]`
11	$[-X+Y, Y, -Z]$	$[x, 2x, 0]$	`[11]`
12	$[X, X-Y, -Z]$	$[x, -x, 0]$	`[12]`

continued ...

Table 81

No.	vector	center	mapping
13	$[-X, -Y, -Z]$	$[-x, -2x, 0]$	`[13]`
14	$[Y, -X+Y, -Z]$	$[2x, x, 0]$	`[14]`
15	$[X-Y, X, -Z]$	$[-x, x, 0]$	`[15]`
16	$[X, Y, -Z]$	$[x, 2x, 0]$	`[16]`
17	$[-Y, X-Y, -Z]$	$[-2x, -x, 0]$	`[17]`
18	$[-X+Y, -X, -Z]$	$[x, -x, 0]$	`[18]`
19	$[-Y, -X, Z]$	$[-2x, -x, 0]$	`[19]`
20	$[-X+Y, Y, Z]$	$[x, 2x, 0]$	`[20]`
21	$[X, X-Y, Z]$	$[x, -x, 0]$	`[21]`
22	$[Y, X, Z]$	$[2x, x, 0]$	`[22]`
23	$[X-Y, -Y, Z]$	$[-x, -2x, 0]$	`[23]`
24	$[-X, -X+Y, Z]$	$[-x, x, 0]$	`[24]`

* Wyckoff site: `6m`, site symmetry: `mm2`

Table 82: Wyckoff bond: `6a@6m`, bond symmetry: `mm2`

No.	vector	center	mapping
1	$[X, 2X, 0]$	$\left[x, 2x, \frac{1}{2}\right]$	`[1,11,16,20]`
2	$[-2X, -X, 0]$	$\left[-2x, -x, \frac{1}{2}\right]$	`[2,10,17,19]`
3	$[X, -X, 0]$	$\left[x, -x, \frac{1}{2}\right]$	`[3,12,18,21]`
4	$[-X, -2X, 0]$	$\left[-x, -2x, \frac{1}{2}\right]$	`[4,8,13,23]`
5	$[2X, X, 0]$	$\left[2x, x, \frac{1}{2}\right]$	`[5,7,14,22]`
6	$[-X, X, 0]$	$\left[-x, x, \frac{1}{2}\right]$	`[6,9,15,24]`

Table 83: Wyckoff bond: `6b@6m`, bond symmetry: `mm2`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[x, 2x, \frac{1}{2}\right]$	`[1,-11,16,-20]`
2	$[0, X, 0]$	$\left[-2x, -x, \frac{1}{2}\right]$	`[2,-10,17,-19]`
3	$[-X, -X, 0]$	$\left[x, -x, \frac{1}{2}\right]$	`[3,-12,18,-21]`
4	$[-X, 0, 0]$	$\left[-x, -2x, \frac{1}{2}\right]$	`[4,-8,13,-23]`
5	$[0, -X, 0]$	$\left[2x, x, \frac{1}{2}\right]$	`[5,-7,14,-22]`
6	$[X, X, 0]$	$\left[-x, x, \frac{1}{2}\right]$	`[6,-9,15,-24]`

Table 84: Wyckoff bond: `6c@6m`, bond symmetry: `mm2`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[x, 2x, \frac{1}{2}\right]$	`[1,-11,-16,20]`
2	$[0, 0, Z]$	$\left[-2x, -x, \frac{1}{2}\right]$	`[2,-10,-17,19]`
3	$[0, 0, Z]$	$\left[x, -x, \frac{1}{2}\right]$	`[3,-12,-18,21]`
4	$[0, 0, Z]$	$\left[-x, -2x, \frac{1}{2}\right]$	`[4,-8,-13,23]`
5	$[0, 0, Z]$	$\left[2x, x, \frac{1}{2}\right]$	`[5,-7,-14,22]`
6	$[0, 0, Z]$	$\left[-x, x, \frac{1}{2}\right]$	`[6,-9,-15,24]`

Table 85: Wyckoff bond: `12d@6m`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X,\ Y,\ 0]$	$\left[x,\ 2x,\ \frac{1}{2}\right]$	`[1,16]`
2	$[-Y,\ X-Y,\ 0]$	$\left[-2x,\ -x,\ \frac{1}{2}\right]$	`[2,17]`
3	$[-X+Y,\ -X,\ 0]$	$\left[x,\ -x,\ \frac{1}{2}\right]$	`[3,18]`
4	$[-X,\ -Y,\ 0]$	$\left[-x,\ -2x,\ \frac{1}{2}\right]$	`[4,13]`
5	$[Y,\ -X+Y,\ 0]$	$\left[2x,\ x,\ \frac{1}{2}\right]$	`[5,14]`
6	$[X-Y,\ X,\ 0]$	$\left[-x,\ x,\ \frac{1}{2}\right]$	`[6,15]`
7	$[Y,\ X,\ 0]$	$\left[2x,\ x,\ \frac{1}{2}\right]$	`[7,22]`
8	$[X-Y,\ -Y,\ 0]$	$\left[-x,\ -2x,\ \frac{1}{2}\right]$	`[8,23]`
9	$[-X,\ -X+Y,\ 0]$	$\left[-x,\ x,\ \frac{1}{2}\right]$	`[9,24]`
10	$[-Y,\ -X,\ 0]$	$\left[-2x,\ -x,\ \frac{1}{2}\right]$	`[10,19]`
11	$[-X+Y,\ Y,\ 0]$	$\left[x,\ 2x,\ \frac{1}{2}\right]$	`[11,20]`
12	$[X,\ X-Y,\ 0]$	$\left[x,\ -x,\ \frac{1}{2}\right]$	`[12,21]`

Table 86: Wyckoff bond: `12e@6m`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X,\ 2X,\ Z]$	$\left[x,\ 2x,\ \frac{1}{2}\right]$	`[1,20]`
2	$[-2X,\ -X,\ Z]$	$\left[-2x,\ -x,\ \frac{1}{2}\right]$	`[2,19]`
3	$[X,\ -X,\ Z]$	$\left[x,\ -x,\ \frac{1}{2}\right]$	`[3,21]`
4	$[-X,\ -2X,\ Z]$	$\left[-x,\ -2x,\ \frac{1}{2}\right]$	`[4,23]`
5	$[2X,\ X,\ Z]$	$\left[2x,\ x,\ \frac{1}{2}\right]$	`[5,22]`
6	$[-X,\ X,\ Z]$	$\left[-x,\ x,\ \frac{1}{2}\right]$	`[6,24]`
7	$[2X,\ X,\ -Z]$	$\left[2x,\ x,\ \frac{1}{2}\right]$	`[7,14]`
8	$[-X,\ -2X,\ -Z]$	$\left[-x,\ -2x,\ \frac{1}{2}\right]$	`[8,13]`
9	$[-X,\ X,\ -Z]$	$\left[-x,\ x,\ \frac{1}{2}\right]$	`[9,15]`
10	$[-2X,\ -X,\ -Z]$	$\left[-2x,\ -x,\ \frac{1}{2}\right]$	`[10,17]`
11	$[X,\ 2X,\ -Z]$	$\left[x,\ 2x,\ \frac{1}{2}\right]$	`[11,16]`
12	$[X,\ -X,\ -Z]$	$\left[x,\ -x,\ \frac{1}{2}\right]$	`[12,18]`

Table 87: Wyckoff bond: `12f@6m`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\ 0,\ Z]$	$\left[x,\ 2x,\ \frac{1}{2}\right]$	`[1,-11]`
2	$[0,\ X,\ Z]$	$\left[-2x,\ -x,\ \frac{1}{2}\right]$	`[2,-10]`
3	$[-X,\ -X,\ Z]$	$\left[x,\ -x,\ \frac{1}{2}\right]$	`[3,-12]`
4	$[-X,\ 0,\ Z]$	$\left[-x,\ -2x,\ \frac{1}{2}\right]$	`[4,-8]`
5	$[0,\ -X,\ Z]$	$\left[2x,\ x,\ \frac{1}{2}\right]$	`[5,-7]`
6	$[X,\ X,\ Z]$	$\left[-x,\ x,\ \frac{1}{2}\right]$	`[6,-9]`
7	$[-X,\ 0,\ -Z]$	$\left[-x,\ -2x,\ \frac{1}{2}\right]$	`[13,-23]`
8	$[0,\ -X,\ -Z]$	$\left[2x,\ x,\ \frac{1}{2}\right]$	`[14,-22]`
9	$[X,\ X,\ -Z]$	$\left[-x,\ x,\ \frac{1}{2}\right]$	`[15,-24]`
10	$[X,\ 0,\ -Z]$	$\left[x,\ 2x,\ \frac{1}{2}\right]$	`[16,-20]`
11	$[0,\ X,\ -Z]$	$\left[-2x,\ -x,\ \frac{1}{2}\right]$	`[17,-19]`
12	$[-X,\ -X,\ -Z]$	$\left[x,\ -x,\ \frac{1}{2}\right]$	`[18,-21]`

Table 88: Wyckoff bond: `24g@6m`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, 2x, \frac{1}{2}\right]$	`[1]`
2	$[-Y, X-Y, Z]$	$\left[-2x, -x, \frac{1}{2}\right]$	`[2]`
3	$[-X+Y, -X, Z]$	$\left[x, -x, \frac{1}{2}\right]$	`[3]`
4	$[-X, -Y, Z]$	$\left[-x, -2x, \frac{1}{2}\right]$	`[4]`
5	$[Y, -X+Y, Z]$	$\left[2x, x, \frac{1}{2}\right]$	`[5]`
6	$[X-Y, X, Z]$	$\left[-x, x, \frac{1}{2}\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[2x, x, \frac{1}{2}\right]$	`[7]`
8	$[X-Y, -Y, -Z]$	$\left[-x, -2x, \frac{1}{2}\right]$	`[8]`
9	$[-X, -X+Y, -Z]$	$\left[-x, x, \frac{1}{2}\right]$	`[9]`
10	$[-Y, -X, -Z]$	$\left[-2x, -x, \frac{1}{2}\right]$	`[10]`
11	$[-X+Y, Y, -Z]$	$\left[x, 2x, \frac{1}{2}\right]$	`[11]`
12	$[X, X-Y, -Z]$	$\left[x, -x, \frac{1}{2}\right]$	`[12]`
13	$[-X, -Y, -Z]$	$\left[-x, -2x, \frac{1}{2}\right]$	`[13]`
14	$[Y, -X+Y, -Z]$	$\left[2x, x, \frac{1}{2}\right]$	`[14]`
15	$[X-Y, X, -Z]$	$\left[-x, x, \frac{1}{2}\right]$	`[15]`
16	$[X, Y, -Z]$	$\left[x, 2x, \frac{1}{2}\right]$	`[16]`
17	$[-Y, X-Y, -Z]$	$\left[-2x, -x, \frac{1}{2}\right]$	`[17]`
18	$[-X+Y, -X, -Z]$	$\left[x, -x, \frac{1}{2}\right]$	`[18]`
19	$[-Y, -X, Z]$	$\left[-2x, -x, \frac{1}{2}\right]$	`[19]`
20	$[-X+Y, Y, Z]$	$\left[x, 2x, \frac{1}{2}\right]$	`[20]`
21	$[X, X-Y, Z]$	$\left[x, -x, \frac{1}{2}\right]$	`[21]`
22	$[Y, X, Z]$	$\left[2x, x, \frac{1}{2}\right]$	`[22]`
23	$[X-Y, -Y, Z]$	$\left[-x, -2x, \frac{1}{2}\right]$	`[23]`
24	$[-X, -X+Y, Z]$	$\left[-x, x, \frac{1}{2}\right]$	`[24]`

* Wyckoff site: `12n`, site symmetry: `..m`

Table 89: Wyckoff bond: `12a@12n`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, 0, Z]$	$[x, 0, z]$	`[1,23]`
2	$[0, X, Z]$	$[0, x, z]$	`[2,22]`
3	$[-X, -X, Z]$	$[-x, -x, z]$	`[3,24]`
4	$[-X, 0, Z]$	$[-x, 0, z]$	`[4,20]`
5	$[0, -X, Z]$	$[0, -x, z]$	`[5,19]`
6	$[X, X, Z]$	$[x, x, z]$	`[6,21]`
7	$[0, X, -Z]$	$[0, x, -z]$	`[7,17]`
8	$[X, 0, -Z]$	$[x, 0, -z]$	`[8,16]`
9	$[-X, -X, -Z]$	$[-x, -x, -z]$	`[9,18]`
10	$[0, -X, -Z]$	$[0, -x, -z]$	`[10,14]`
11	$[-X, 0, -Z]$	$[-x, 0, -z]$	`[11,13]`
12	$[X, X, -Z]$	$[x, x, -z]$	`[12,15]`

Table 90: Wyckoff bond: `12b@12n`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, 2X, 0]$	$[x, 0, z]$	`[1,-23]`
2	$[-2X, -X, 0]$	$[0, x, z]$	`[2,-22]`
3	$[X, -X, 0]$	$[-x, -x, z]$	`[3,-24]`
4	$[-X, -2X, 0]$	$[-x, 0, z]$	`[4,-20]`
5	$[2X, X, 0]$	$[0, -x, z]$	`[5,-19]`
6	$[-X, X, 0]$	$[x, x, z]$	`[6,-21]`
7	$[2X, X, 0]$	$[0, x, -z]$	`[7,-17]`
8	$[-X, -2X, 0]$	$[x, 0, -z]$	`[8,-16]`
9	$[-X, X, 0]$	$[-x, -x, -z]$	`[9,-18]`
10	$[-2X, -X, 0]$	$[0, -x, -z]$	`[10,-14]`
11	$[X, 2X, 0]$	$[-x, 0, -z]$	`[11,-13]`
12	$[X, -X, 0]$	$[x, x, -z]$	`[12,-15]`

Table 91: Wyckoff bond: `24c@12n`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, 0, z]$	`[1]`
2	$[-Y, X-Y, Z]$	$[0, x, z]$	`[2]`
3	$[-X+Y, -X, Z]$	$[-x, -x, z]$	`[3]`
4	$[-X, -Y, Z]$	$[-x, 0, z]$	`[4]`
5	$[Y, -X+Y, Z]$	$[0, -x, z]$	`[5]`
6	$[X-Y, X, Z]$	$[x, x, z]$	`[6]`
7	$[Y, X, -Z]$	$[0, x, -z]$	`[7]`
8	$[X-Y, -Y, -Z]$	$[x, 0, -z]$	`[8]`
9	$[-X, -X+Y, -Z]$	$[-x, -x, -z]$	`[9]`
10	$[-Y, -X, -Z]$	$[0, -x, -z]$	`[10]`
11	$[-X+Y, Y, -Z]$	$[-x, 0, -z]$	`[11]`
12	$[X, X-Y, -Z]$	$[x, x, -z]$	`[12]`
13	$[-X, -Y, -Z]$	$[-x, 0, -z]$	`[13]`
14	$[Y, -X+Y, -Z]$	$[0, -x, -z]$	`[14]`
15	$[X-Y, X, -Z]$	$[x, x, -z]$	`[15]`
16	$[X, Y, -Z]$	$[x, 0, -z]$	`[16]`
17	$[-Y, X-Y, -Z]$	$[0, x, -z]$	`[17]`
18	$[-X+Y, -X, -Z]$	$[-x, -x, -z]$	`[18]`
19	$[-Y, -X, Z]$	$[0, -x, z]$	`[19]`
20	$[-X+Y, Y, Z]$	$[-x, 0, z]$	`[20]`
21	$[X, X-Y, Z]$	$[x, x, z]$	`[21]`
22	$[Y, X, Z]$	$[0, x, z]$	`[22]`
23	$[X-Y, -Y, Z]$	$[x, 0, z]$	`[23]`
24	$[-X, -X+Y, Z]$	$[-x, -x, z]$	`[24]`

* Wyckoff site: `12o`, site symmetry: `.m.`

Table 92: Wyckoff bond: `12a@12o`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X, 2X, Z]$	$[x, 2x, z]$	`[1,20]`
2	$[-2X, -X, Z]$	$[-2x, -x, z]$	`[2,19]`
3	$[X, -X, Z]$	$[x, -x, z]$	`[3,21]`
4	$[-X, -2X, Z]$	$[-x, -2x, z]$	`[4,23]`
5	$[2X, X, Z]$	$[2x, x, z]$	`[5,22]`
6	$[-X, X, Z]$	$[-x, x, z]$	`[6,24]`
7	$[2X, X, -Z]$	$[2x, x, -z]$	`[7,14]`
8	$[-X, -2X, -Z]$	$[-x, -2x, -z]$	`[8,13]`
9	$[-X, X, -Z]$	$[-x, x, -z]$	`[9,15]`
10	$[-2X, -X, -Z]$	$[-2x, -x, -z]$	`[10,17]`
11	$[X, 2X, -Z]$	$[x, 2x, -z]$	`[11,16]`
12	$[X, -X, -Z]$	$[x, -x, -z]$	`[12,18]`

Table 93: Wyckoff bond: `12b@12o`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[x, 2x, z]$	`[1,-20]`
2	$[0, X, 0]$	$[-2x, -x, z]$	`[2,-19]`
3	$[-X, -X, 0]$	$[x, -x, z]$	`[3,-21]`
4	$[-X, 0, 0]$	$[-x, -2x, z]$	`[4,-23]`
5	$[0, -X, 0]$	$[2x, x, z]$	`[5,-22]`
6	$[X, X, 0]$	$[-x, x, z]$	`[6,-24]`
7	$[0, X, 0]$	$[2x, x, -z]$	`[7,-14]`
8	$[X, 0, 0]$	$[-x, -2x, -z]$	`[8,-13]`
9	$[-X, -X, 0]$	$[-x, x, -z]$	`[9,-15]`
10	$[0, -X, 0]$	$[-2x, -x, -z]$	`[10,-17]`
11	$[-X, 0, 0]$	$[x, 2x, -z]$	`[11,-16]`
12	$[X, X, 0]$	$[x, -x, -z]$	`[12,-18]`

Table 94: Wyckoff bond: `24c@12o`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, 2x, z]$	`[1]`
2	$[-Y, X-Y, Z]$	$[-2x, -x, z]$	`[2]`
3	$[-X+Y, -X, Z]$	$[x, -x, z]$	`[3]`
4	$[-X, -Y, Z]$	$[-x, -2x, z]$	`[4]`
5	$[Y, -X+Y, Z]$	$[2x, x, z]$	`[5]`
6	$[X-Y, X, Z]$	$[-x, x, z]$	`[6]`
7	$[Y, X, -Z]$	$[2x, x, -z]$	`[7]`
8	$[X-Y, -Y, -Z]$	$[-x, -2x, -z]$	`[8]`
9	$[-X, -X+Y, -Z]$	$[-x, x, -z]$	`[9]`
10	$[-Y, -X, -Z]$	$[-2x, -x, -z]$	`[10]`
11	$[-X+Y, Y, -Z]$	$[x, 2x, -z]$	`[11]`
12	$[X, X-Y, -Z]$	$[x, -x, -z]$	`[12]`

continued ...

Table 94

No.	vector	center	mapping
13	$[-X, -Y, -Z]$	$[-x, -2x, -z]$	[13]
14	$[Y, -X+Y, -Z]$	$[2x, x, -z]$	[14]
15	$[X-Y, X, -Z]$	$[-x, x, -z]$	[15]
16	$[X, Y, -Z]$	$[x, 2x, -z]$	[16]
17	$[-Y, X-Y, -Z]$	$[-2x, -x, -z]$	[17]
18	$[-X+Y, -X, -Z]$	$[x, -x, -z]$	[18]
19	$[-Y, -X, Z]$	$[-2x, -x, z]$	[19]
20	$[-X+Y, Y, Z]$	$[x, 2x, z]$	[20]
21	$[X, X-Y, Z]$	$[x, -x, z]$	[21]
22	$[Y, X, Z]$	$[2x, x, z]$	[22]
23	$[X-Y, -Y, Z]$	$[-x, -2x, z]$	[23]
24	$[-X, -X+Y, Z]$	$[-x, x, z]$	[24]

* Wyckoff site: 12p, site symmetry: m..

Table 95: Wyckoff bond: 12a@12p, bond symmetry: m..

No.	vector	center	mapping
1	$[X, Y, 0]$	$[x, y, 0]$	[1,16]
2	$[-Y, X-Y, 0]$	$[-y, x-y, 0]$	[2,17]
3	$[-X+Y, -X, 0]$	$[-x+y, -x, 0]$	[3,18]
4	$[-X, -Y, 0]$	$[-x, -y, 0]$	[4,13]
5	$[Y, -X+Y, 0]$	$[y, -x+y, 0]$	[5,14]
6	$[X-Y, X, 0]$	$[x-y, x, 0]$	[6,15]
7	$[Y, X, 0]$	$[y, x, 0]$	[7,22]
8	$[X-Y, -Y, 0]$	$[x-y, -y, 0]$	[8,23]
9	$[-X, -X+Y, 0]$	$[-x, -x+y, 0]$	[9,24]
10	$[-Y, -X, 0]$	$[-y, -x, 0]$	[10,19]
11	$[-X+Y, Y, 0]$	$[-x+y, y, 0]$	[11,20]
12	$[X, X-Y, 0]$	$[x, x-y, 0]$	[12,21]

Table 96: Wyckoff bond: 12b@12p, bond symmetry: m..

No.	vector	center	mapping
1	$[0, 0, Z]$	$[x, y, 0]$	[1,-16]
2	$[0, 0, Z]$	$[-y, x-y, 0]$	[2,-17]
3	$[0, 0, Z]$	$[-x+y, -x, 0]$	[3,-18]
4	$[0, 0, Z]$	$[-x, -y, 0]$	[4,-13]
5	$[0, 0, Z]$	$[y, -x+y, 0]$	[5,-14]
6	$[0, 0, Z]$	$[x-y, x, 0]$	[6,-15]
7	$[0, 0, -Z]$	$[y, x, 0]$	[7,-22]
8	$[0, 0, -Z]$	$[x-y, -y, 0]$	[8,-23]
9	$[0, 0, -Z]$	$[-x, -x+y, 0]$	[9,-24]
10	$[0, 0, -Z]$	$[-y, -x, 0]$	[10,-19]
11	$[0, 0, -Z]$	$[-x+y, y, 0]$	[11,-20]
12	$[0, 0, -Z]$	$[x, x-y, 0]$	[12,-21]

Table 97: Wyckoff bond: `24c@12p`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, 0]$	`[1]`
2	$[-Y, X-Y, Z]$	$[-y, x-y, 0]$	`[2]`
3	$[-X+Y, -X, Z]$	$[-x+y, -x, 0]$	`[3]`
4	$[-X, -Y, Z]$	$[-x, -y, 0]$	`[4]`
5	$[Y, -X+Y, Z]$	$[y, -x+y, 0]$	`[5]`
6	$[X-Y, X, Z]$	$[x-y, x, 0]$	`[6]`
7	$[Y, X, -Z]$	$[y, x, 0]$	`[7]`
8	$[X-Y, -Y, -Z]$	$[x-y, -y, 0]$	`[8]`
9	$[-X, -X+Y, -Z]$	$[-x, -x+y, 0]$	`[9]`
10	$[-Y, -X, -Z]$	$[-y, -x, 0]$	`[10]`
11	$[-X+Y, Y, -Z]$	$[-x+y, y, 0]$	`[11]`
12	$[X, X-Y, -Z]$	$[x, x-y, 0]$	`[12]`
13	$[-X, -Y, -Z]$	$[-x, -y, 0]$	`[13]`
14	$[Y, -X+Y, -Z]$	$[y, -x+y, 0]$	`[14]`
15	$[X-Y, X, -Z]$	$[x-y, x, 0]$	`[15]`
16	$[X, Y, -Z]$	$[x, y, 0]$	`[16]`
17	$[-Y, X-Y, -Z]$	$[-y, x-y, 0]$	`[17]`
18	$[-X+Y, -X, -Z]$	$[-x+y, -x, 0]$	`[18]`
19	$[-Y, -X, Z]$	$[-y, -x, 0]$	`[19]`
20	$[-X+Y, Y, Z]$	$[-x+y, y, 0]$	`[20]`
21	$[X, X-Y, Z]$	$[x, x-y, 0]$	`[21]`
22	$[Y, X, Z]$	$[y, x, 0]$	`[22]`
23	$[X-Y, -Y, Z]$	$[x-y, -y, 0]$	`[23]`
24	$[-X, -X+Y, Z]$	$[-x, -x+y, 0]$	`[24]`

* Wyckoff site: `12q`, site symmetry: `m..`

Table 98: Wyckoff bond: `12a@12q`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[x, y, \frac{1}{2}\right]$	`[1,16]`
2	$[-Y, X-Y, 0]$	$\left[-y, x-y, \frac{1}{2}\right]$	`[2,17]`
3	$[-X+Y, -X, 0]$	$\left[-x+y, -x, \frac{1}{2}\right]$	`[3,18]`
4	$[-X, -Y, 0]$	$\left[-x, -y, \frac{1}{2}\right]$	`[4,13]`
5	$[Y, -X+Y, 0]$	$\left[y, -x+y, \frac{1}{2}\right]$	`[5,14]`
6	$[X-Y, X, 0]$	$\left[x-y, x, \frac{1}{2}\right]$	`[6,15]`
7	$[Y, X, 0]$	$\left[y, x, \frac{1}{2}\right]$	`[7,22]`
8	$[X-Y, -Y, 0]$	$\left[x-y, -y, \frac{1}{2}\right]$	`[8,23]`
9	$[-X, -X+Y, 0]$	$\left[-x, -x+y, \frac{1}{2}\right]$	`[9,24]`
10	$[-Y, -X, 0]$	$\left[-y, -x, \frac{1}{2}\right]$	`[10,19]`
11	$[-X+Y, Y, 0]$	$\left[-x+y, y, \frac{1}{2}\right]$	`[11,20]`
12	$[X, X-Y, 0]$	$\left[x, x-y, \frac{1}{2}\right]$	`[12,21]`

Table 99: Wyckoff bond: `12b@12q`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[x,\ y,\ \frac{1}{2}\right]$	`[1,-16]`
2	$[0,\ 0,\ Z]$	$\left[-y,\ x-y,\ \frac{1}{2}\right]$	`[2,-17]`
3	$[0,\ 0,\ Z]$	$\left[-x+y,\ -x,\ \frac{1}{2}\right]$	`[3,-18]`
4	$[0,\ 0,\ Z]$	$\left[-x,\ -y,\ \frac{1}{2}\right]$	`[4,-13]`
5	$[0,\ 0,\ Z]$	$\left[y,\ -x+y,\ \frac{1}{2}\right]$	`[5,-14]`
6	$[0,\ 0,\ Z]$	$\left[x-y,\ x,\ \frac{1}{2}\right]$	`[6,-15]`
7	$[0,\ 0,\ -Z]$	$\left[y,\ x,\ \frac{1}{2}\right]$	`[7,-22]`
8	$[0,\ 0,\ -Z]$	$\left[x-y,\ -y,\ \frac{1}{2}\right]$	`[8,-23]`
9	$[0,\ 0,\ -Z]$	$\left[-x,\ -x+y,\ \frac{1}{2}\right]$	`[9,-24]`
10	$[0,\ 0,\ -Z]$	$\left[-y,\ -x,\ \frac{1}{2}\right]$	`[10,-19]`
11	$[0,\ 0,\ -Z]$	$\left[-x+y,\ y,\ \frac{1}{2}\right]$	`[11,-20]`
12	$[0,\ 0,\ -Z]$	$\left[x,\ x-y,\ \frac{1}{2}\right]$	`[12,-21]`

Table 100: Wyckoff bond: `24c@12q`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[x,\ y,\ \frac{1}{2}\right]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$\left[-y,\ x-y,\ \frac{1}{2}\right]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$\left[-x+y,\ -x,\ \frac{1}{2}\right]$	`[3]`
4	$[-X,\ -Y,\ Z]$	$\left[-x,\ -y,\ \frac{1}{2}\right]$	`[4]`
5	$[Y,\ -X+Y,\ Z]$	$\left[y,\ -x+y,\ \frac{1}{2}\right]$	`[5]`
6	$[X-Y,\ X,\ Z]$	$\left[x-y,\ x,\ \frac{1}{2}\right]$	`[6]`
7	$[Y,\ X,\ -Z]$	$\left[y,\ x,\ \frac{1}{2}\right]$	`[7]`
8	$[X-Y,\ -Y,\ -Z]$	$\left[x-y,\ -y,\ \frac{1}{2}\right]$	`[8]`
9	$[-X,\ -X+Y,\ -Z]$	$\left[-x,\ -x+y,\ \frac{1}{2}\right]$	`[9]`
10	$[-Y,\ -X,\ -Z]$	$\left[-y,\ -x,\ \frac{1}{2}\right]$	`[10]`
11	$[-X+Y,\ Y,\ -Z]$	$\left[-x+y,\ y,\ \frac{1}{2}\right]$	`[11]`
12	$[X,\ X-Y,\ -Z]$	$\left[x,\ x-y,\ \frac{1}{2}\right]$	`[12]`
13	$[-X,\ -Y,\ -Z]$	$\left[-x,\ -y,\ \frac{1}{2}\right]$	`[13]`
14	$[Y,\ -X+Y,\ -Z]$	$\left[y,\ -x+y,\ \frac{1}{2}\right]$	`[14]`
15	$[X-Y,\ X,\ -Z]$	$\left[x-y,\ x,\ \frac{1}{2}\right]$	`[15]`
16	$[X,\ Y,\ -Z]$	$\left[x,\ y,\ \frac{1}{2}\right]$	`[16]`
17	$[-Y,\ X-Y,\ -Z]$	$\left[-y,\ x-y,\ \frac{1}{2}\right]$	`[17]`
18	$[-X+Y,\ -X,\ -Z]$	$\left[-x+y,\ -x,\ \frac{1}{2}\right]$	`[18]`
19	$[-Y,\ -X,\ Z]$	$\left[-y,\ -x,\ \frac{1}{2}\right]$	`[19]`
20	$[-X+Y,\ Y,\ Z]$	$\left[-x+y,\ y,\ \frac{1}{2}\right]$	`[20]`
21	$[X,\ X-Y,\ Z]$	$\left[x,\ x-y,\ \frac{1}{2}\right]$	`[21]`
22	$[Y,\ X,\ Z]$	$\left[y,\ x,\ \frac{1}{2}\right]$	`[22]`
23	$[X-Y,\ -Y,\ Z]$	$\left[x-y,\ -y,\ \frac{1}{2}\right]$	`[23]`
24	$[-X,\ -X+Y,\ Z]$	$\left[-x,\ -x+y,\ \frac{1}{2}\right]$	`[24]`

* Wyckoff site: `24r`, site symmetry: `1`

Table 101: Wyckoff bond: `24a@24r`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, z]$	`[1]`
2	$[-Y, X-Y, Z]$	$[-y, x-y, z]$	`[2]`
3	$[-X+Y, -X, Z]$	$[-x+y, -x, z]$	`[3]`
4	$[-X, -Y, Z]$	$[-x, -y, z]$	`[4]`
5	$[Y, -X+Y, Z]$	$[y, -x+y, z]$	`[5]`
6	$[X-Y, X, Z]$	$[x-y, x, z]$	`[6]`
7	$[Y, X, -Z]$	$[y, x, -z]$	`[7]`
8	$[X-Y, -Y, -Z]$	$[x-y, -y, -z]$	`[8]`
9	$[-X, -X+Y, -Z]$	$[-x, -x+y, -z]$	`[9]`
10	$[-Y, -X, -Z]$	$[-y, -x, -z]$	`[10]`
11	$[-X+Y, Y, -Z]$	$[-x+y, y, -z]$	`[11]`
12	$[X, X-Y, -Z]$	$[x, x-y, -z]$	`[12]`
13	$[-X, -Y, -Z]$	$[-x, -y, -z]$	`[13]`
14	$[Y, -X+Y, -Z]$	$[y, -x+y, -z]$	`[14]`
15	$[X-Y, X, -Z]$	$[x-y, x, -z]$	`[15]`
16	$[X, Y, -Z]$	$[x, y, -z]$	`[16]`
17	$[-Y, X-Y, -Z]$	$[-y, x-y, -z]$	`[17]`
18	$[-X+Y, -X, -Z]$	$[-x+y, -x, -z]$	`[18]`
19	$[-Y, -X, Z]$	$[-y, -x, z]$	`[19]`
20	$[-X+Y, Y, Z]$	$[-x+y, y, z]$	`[20]`
21	$[X, X-Y, Z]$	$[x, x-y, z]$	`[21]`
22	$[Y, X, Z]$	$[y, x, z]$	`[22]`
23	$[X-Y, -Y, Z]$	$[x-y, -y, z]$	`[23]`
24	$[-X, -X+Y, Z]$	$[-x, -x+y, z]$	`[24]`